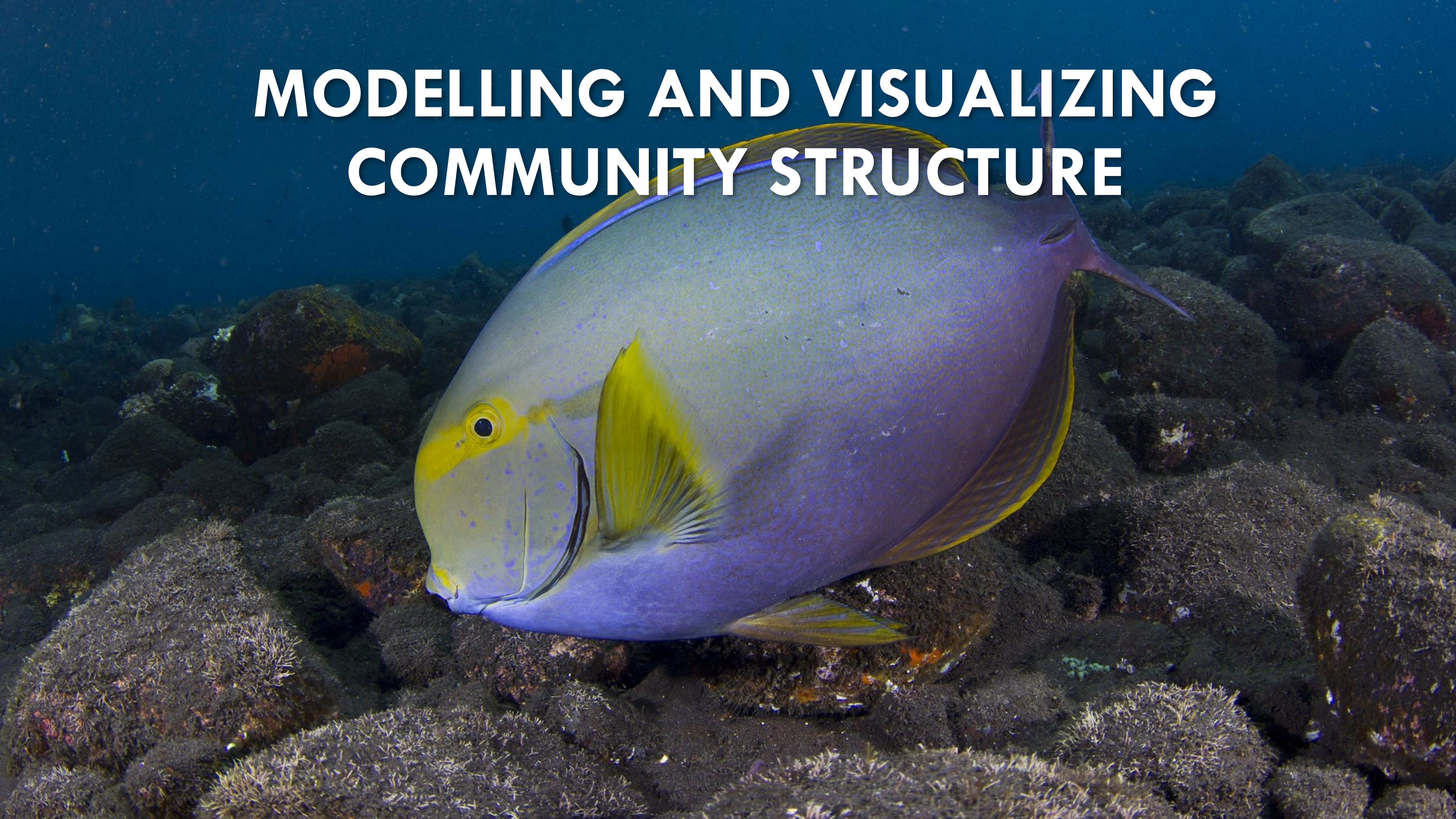
MODELLING AND VISUALIZING COMMUNITY STRUCTURE

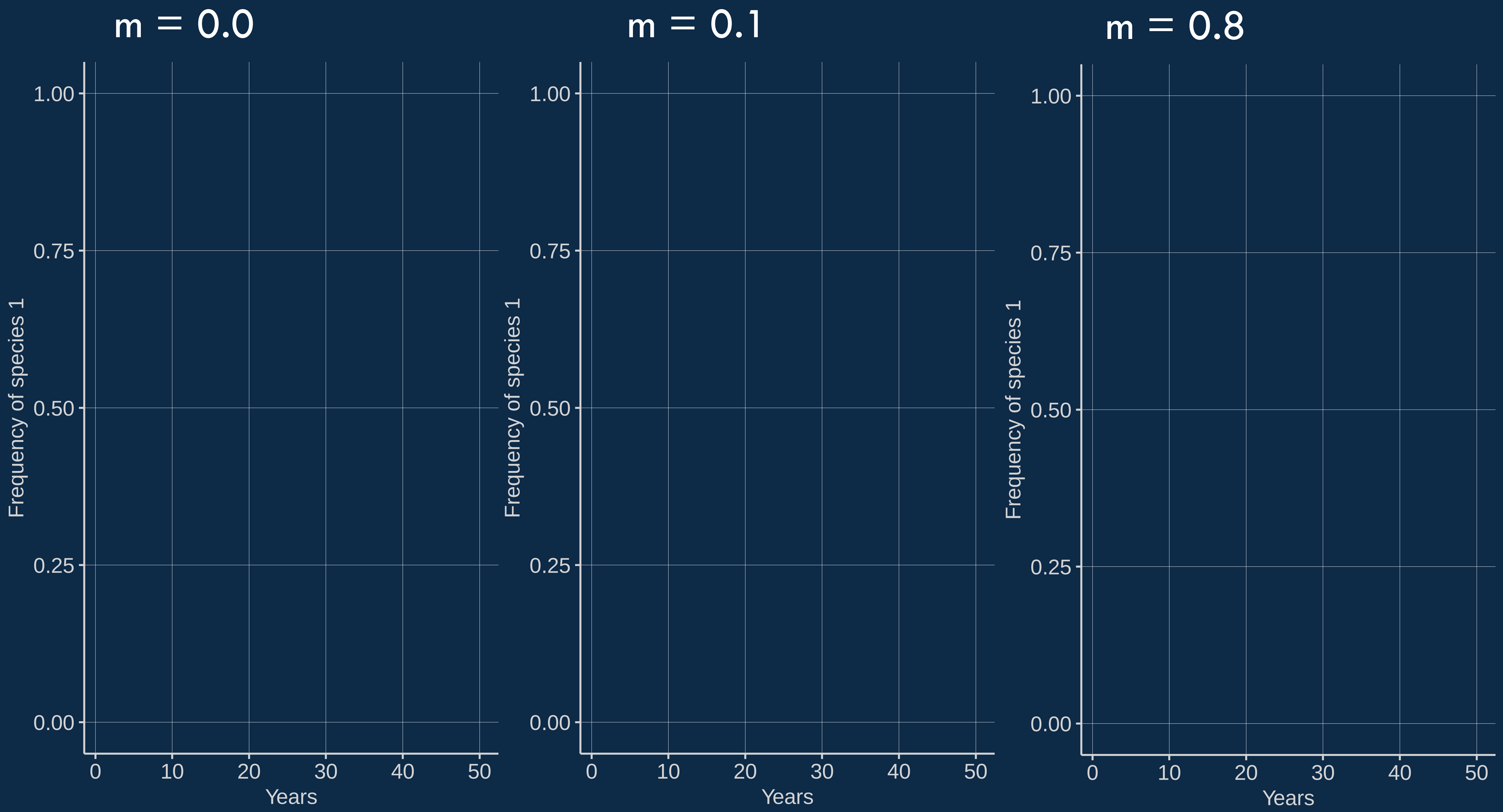


Housekeeping

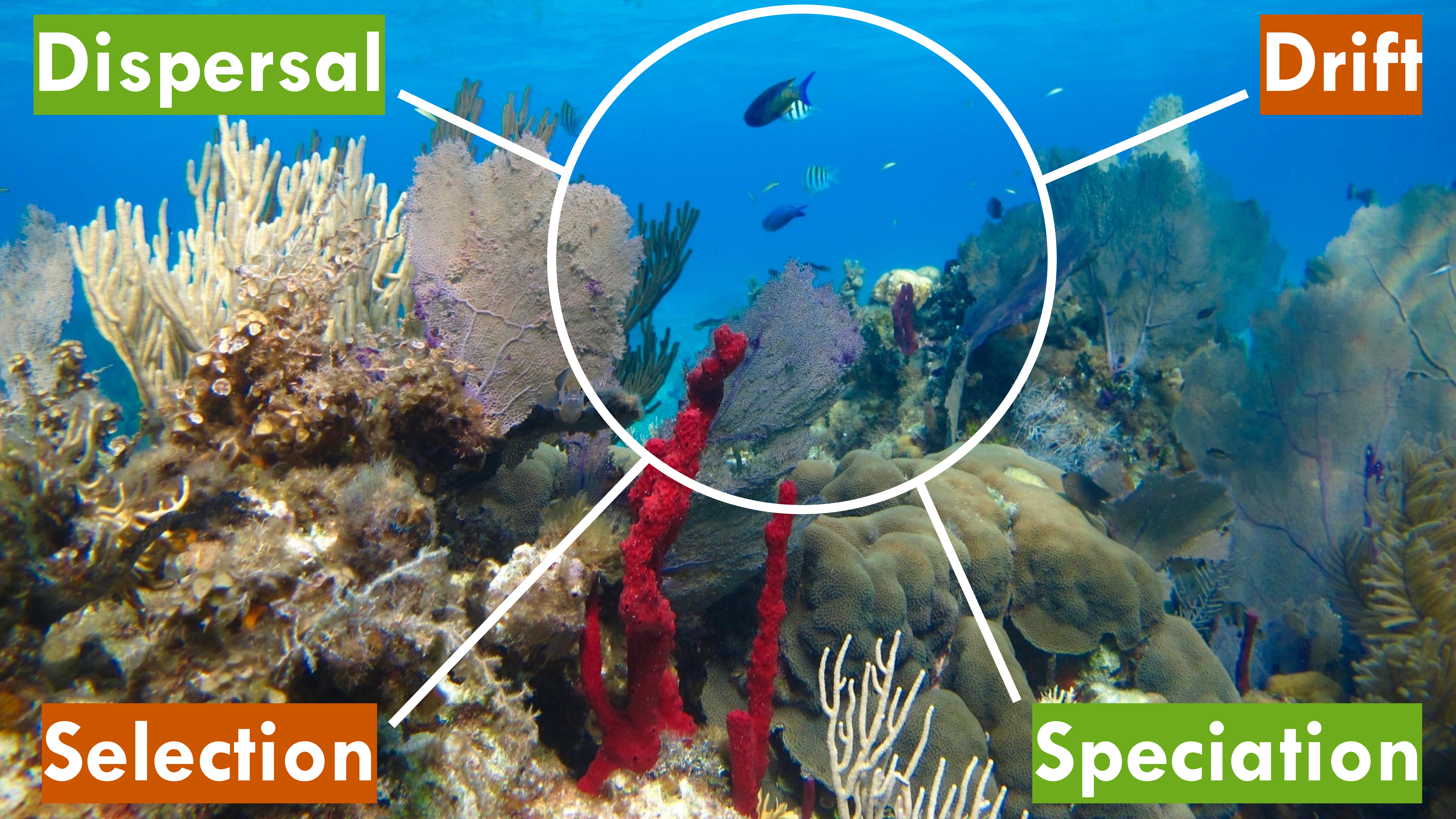
1) Assignment I edits: please resubmit as Word doc

2) More bird surveys on Thursday

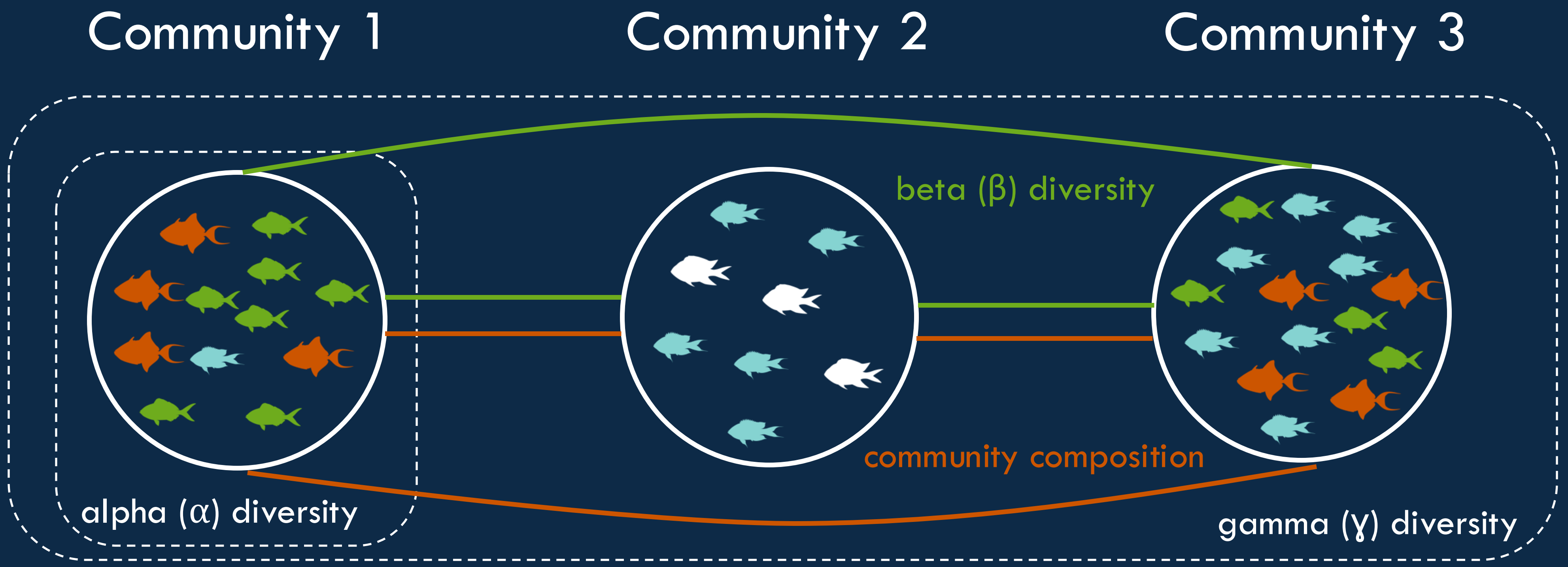

DRIFT
SELECTION
PROOF
DISPERSAL
SPECIATION



m = 0.0
m = 0.1
m = 0.8
Frequency of species 1
1.00
0.75
0.50
0.25
0.00
0
10
20
30
40
50
Years


Dispersal
Drift
Selection
Speciation

First order properties: multiple communities



Cheat sheet

1st order variables, one community:

1) Abundance (integer, non-negative)
2) Species richness (integer, non-negative)
3) Diversity or evenness (decimal, non-negative)
4) Species-abundance distribution (distribution)

1st order variables, many communities:

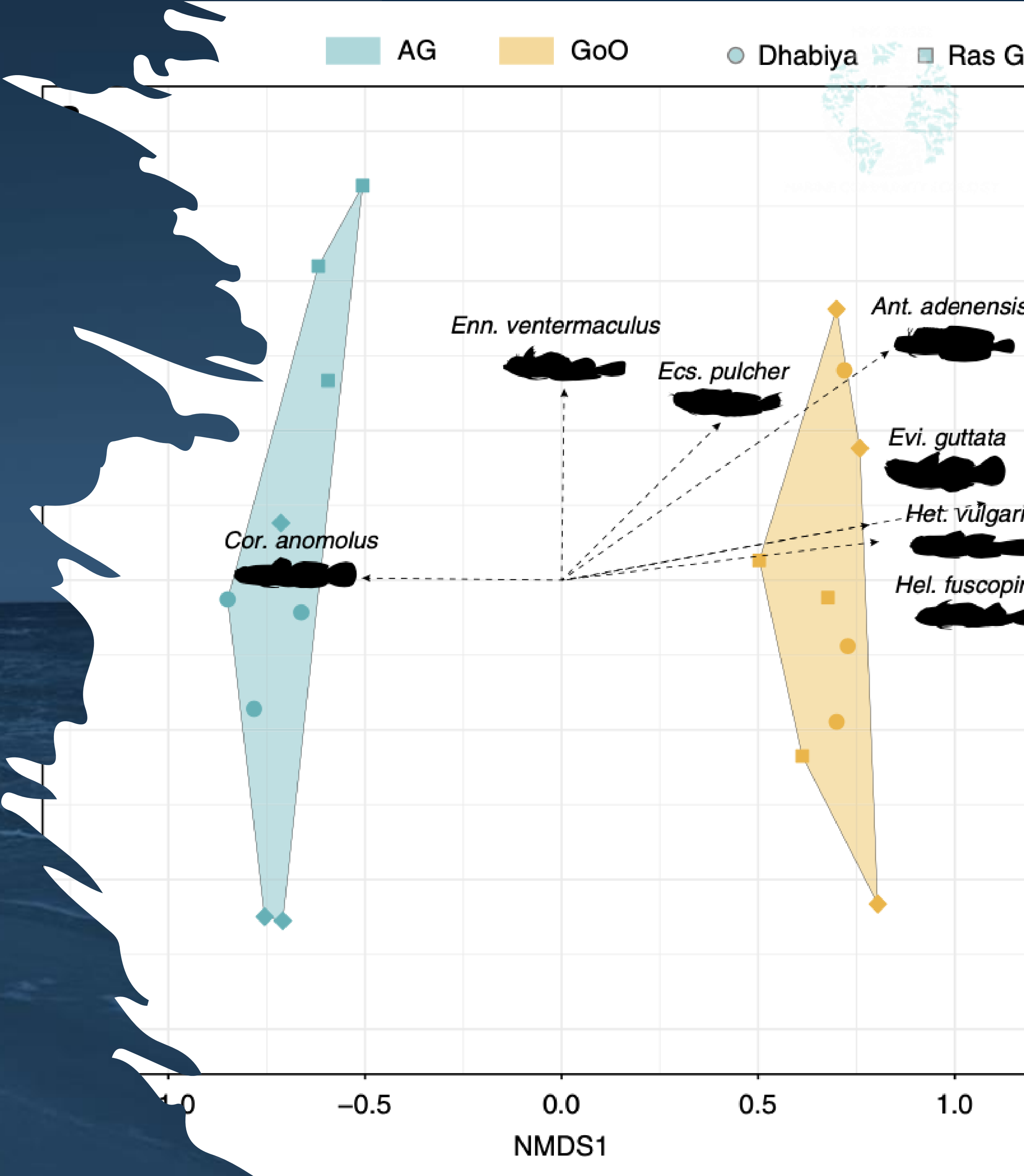
1) Community composition (no value)
2) Beta-diversity (decimal, non-negative)

2nd order variables:

1) Trait diversity and composition (requires species characteristics)
2) Species-environment relationships (requires site characteristics)

COMMUNITY COMPOSITION

ORDINATION


AG
GoO
Dhabiya
Enn. ventermaculus
Ecs. pulcher
Ant. adenensis
Evi. guttata
Cor. anomolus
−0.5
0.0
0.5
1.0
NMDS1

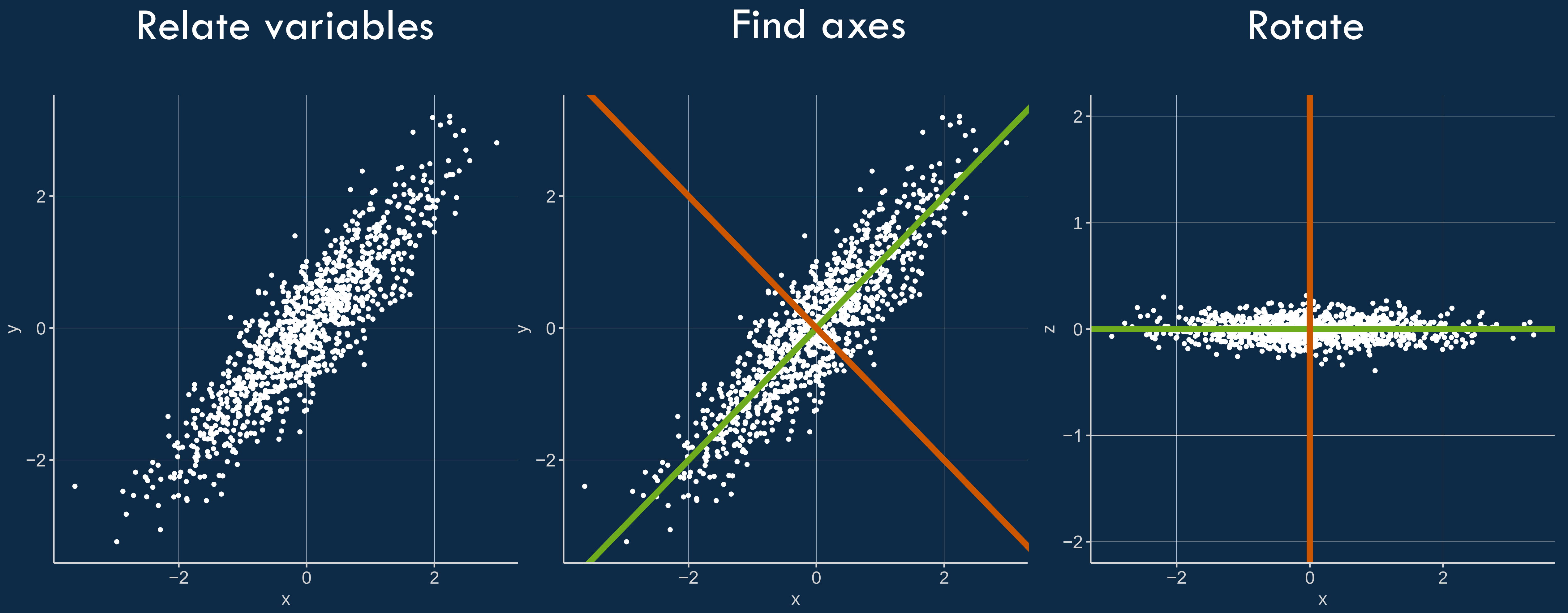
Origin of ordination is the German word “*Ordnung*”

ORDINATION

- Goal of ordination: impose order on messy data
- One of two main intentions:
  - arrange samples along gradients to describe patterns
  - reduce the dimensionality of variation to reflect similarity among sites


"THE DUKE"
Wilson
NFL
COMMISSIONER

AN EXAMPLE



ORDINATION

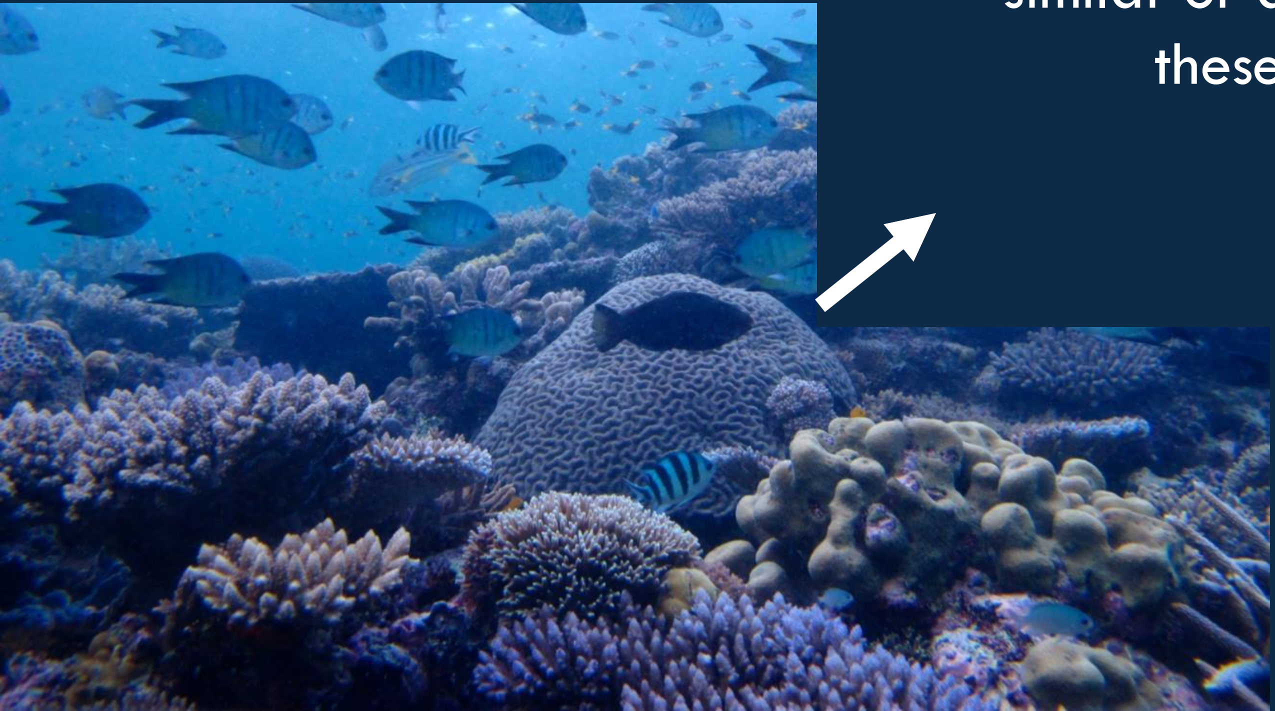
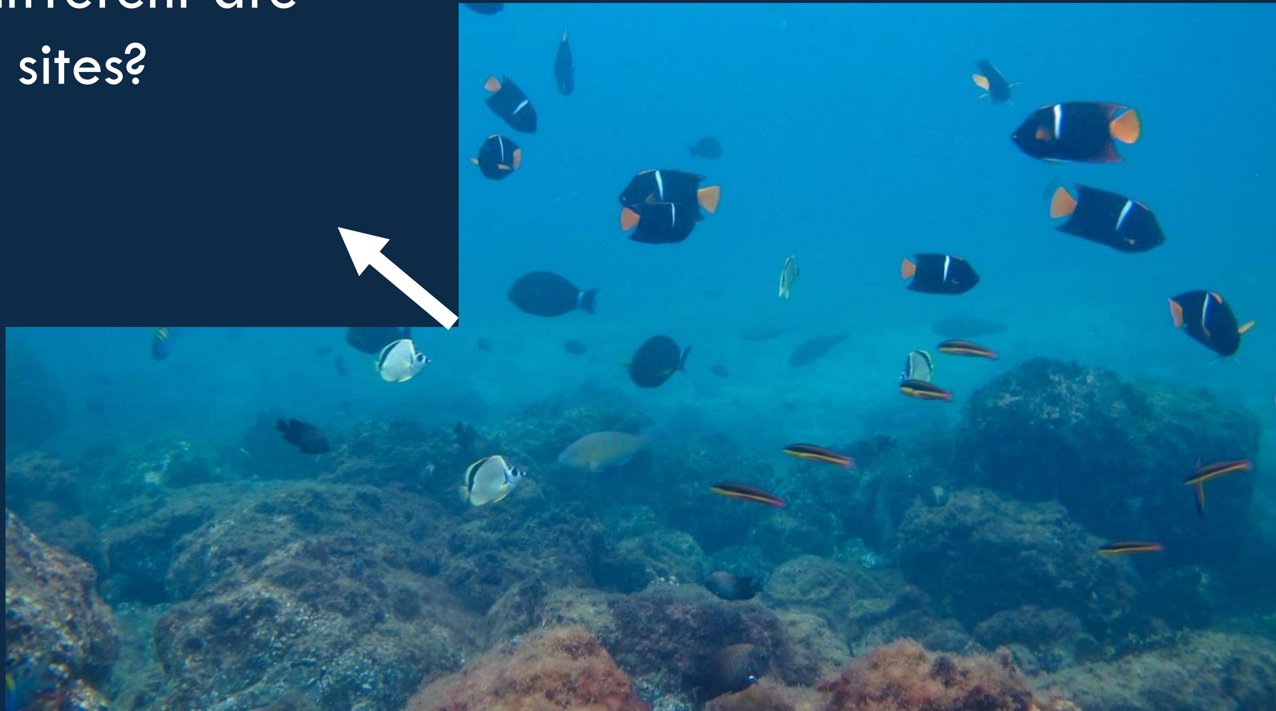
3 FAMILIES:

1) Linear: species have linear distribution across gradient
2) Unimodal: species have non-linear distribution
3) Distance-based: patterns among sites, not species

2 TYPES:

1) Constrained
2) Unconstrained


Based on species, how similar or different are these sites?

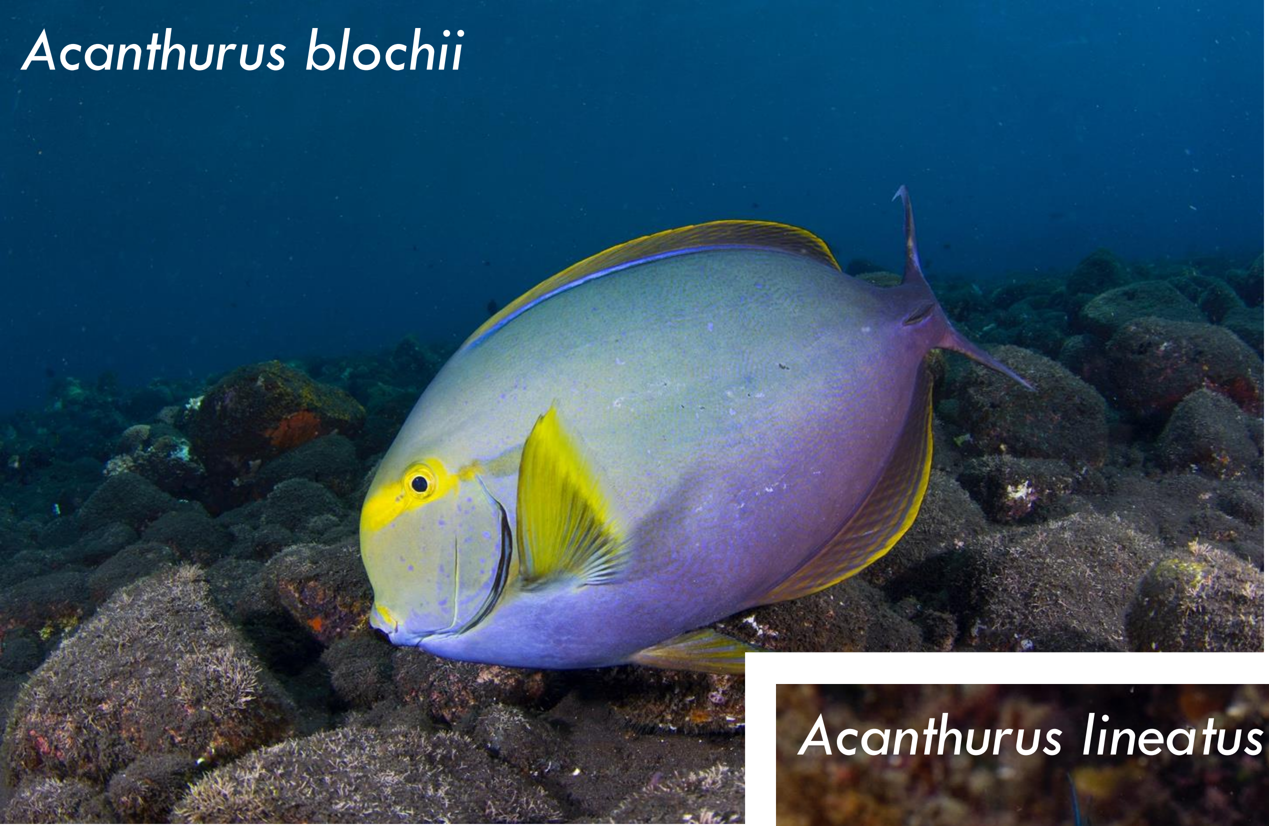



PRINCIPAL COMPONENTS ANALYSIS (PCA)

- linear, unconstrained
- dimensionality-reduction method: maximize information in a small set of variables
- works in four distinct steps

FOUR STEPS OF PCA

- Standardization: scaling and centering
- Compute a covariance matrix between variables
- Compute eigenvectors and eigenvalues of the covariance matrix to identify principal components
- Rotate data along the principal components axes


Acanthurus blochii

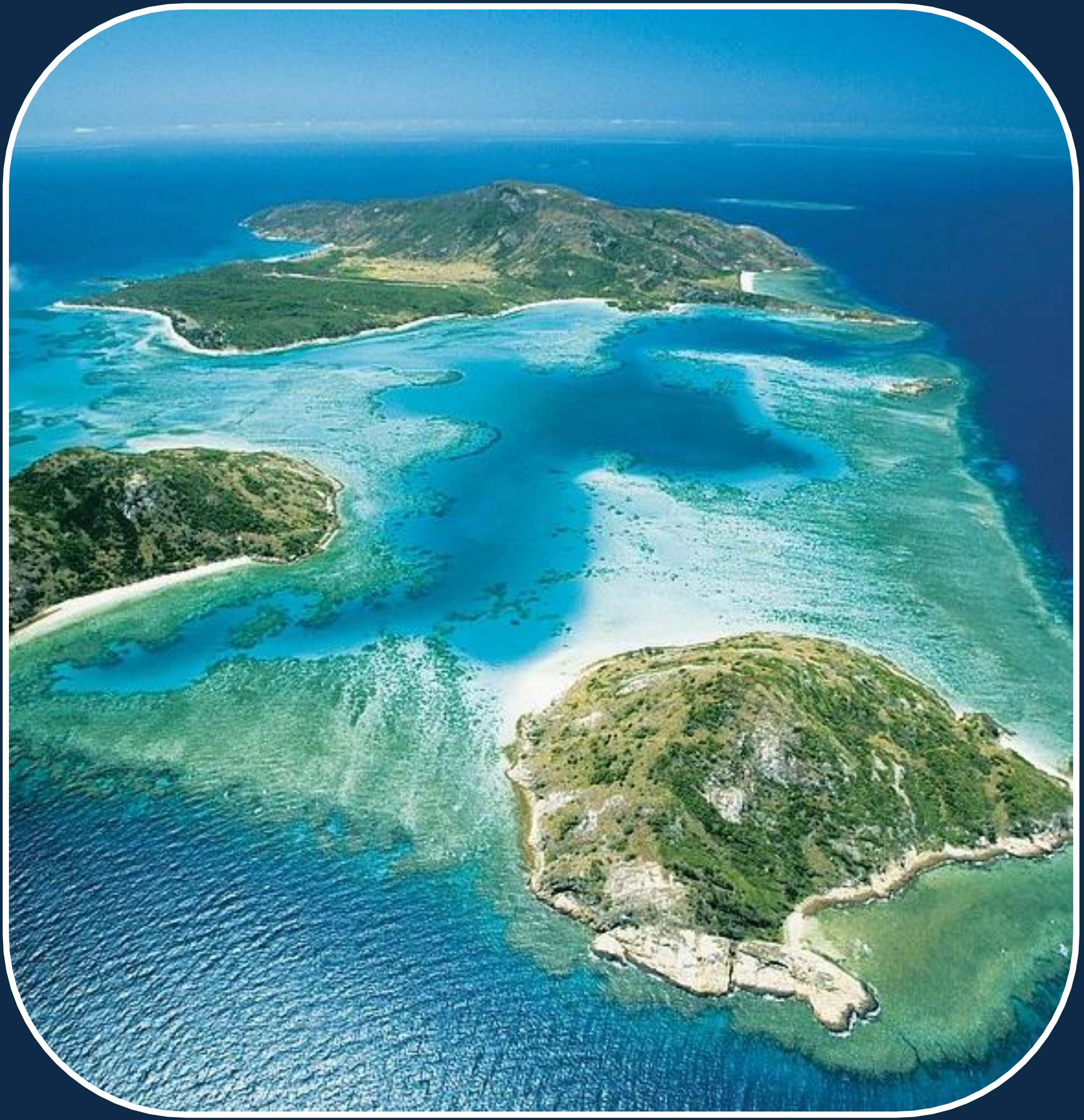
Scarus oviceps

Acanthurus lineatus

Chlorurus microrhinos

Acanthurus nigricans

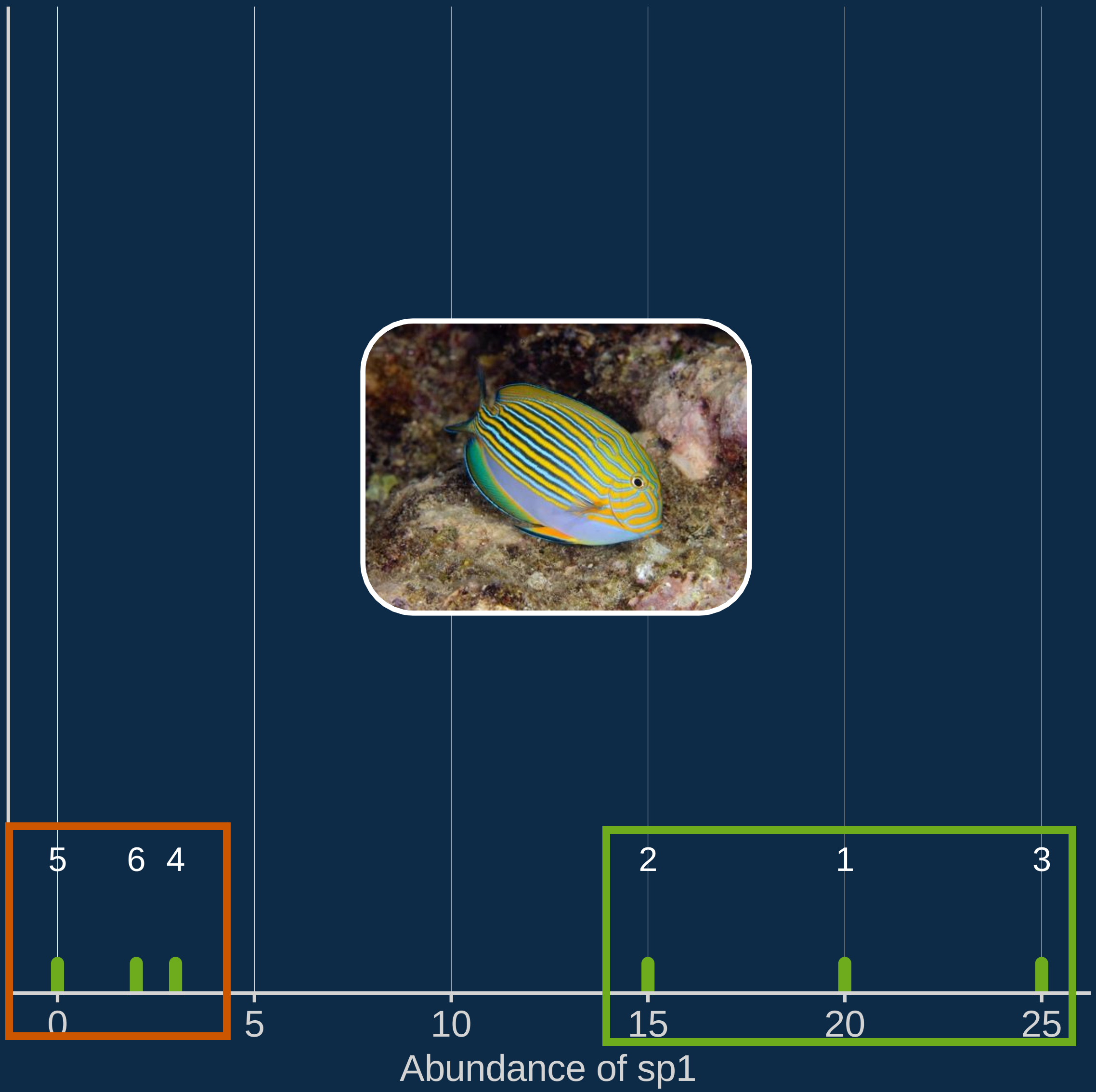

Australia
Lizard Island
Research Station



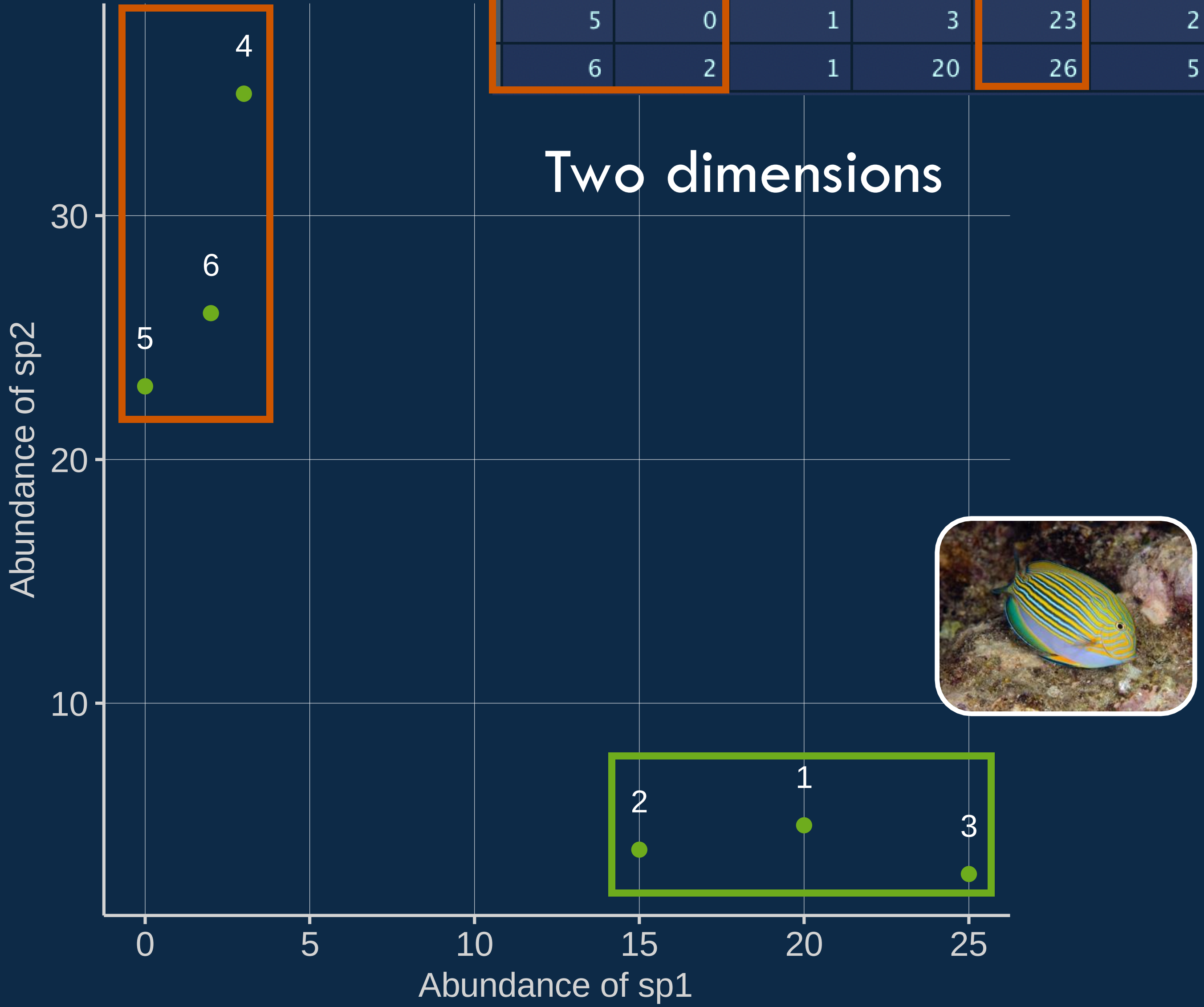
site	alin	cmic	ablo	sovi	anig
1	20	10	1	5	2
2	15	8	2	4	1
3	25	12	2	3	3
4	3	5	33	35	25
5	0	1	3	23	2
6	2	1	20	26	5

site	alin	cmic	ablo	sovi	anig
1	20	10	1	5	2
2	15	8	2	4	1
3	25	12	2	3	3
4	3	5	33	35	25
5	0	1	3	23	2
6	2	1	20	26	5

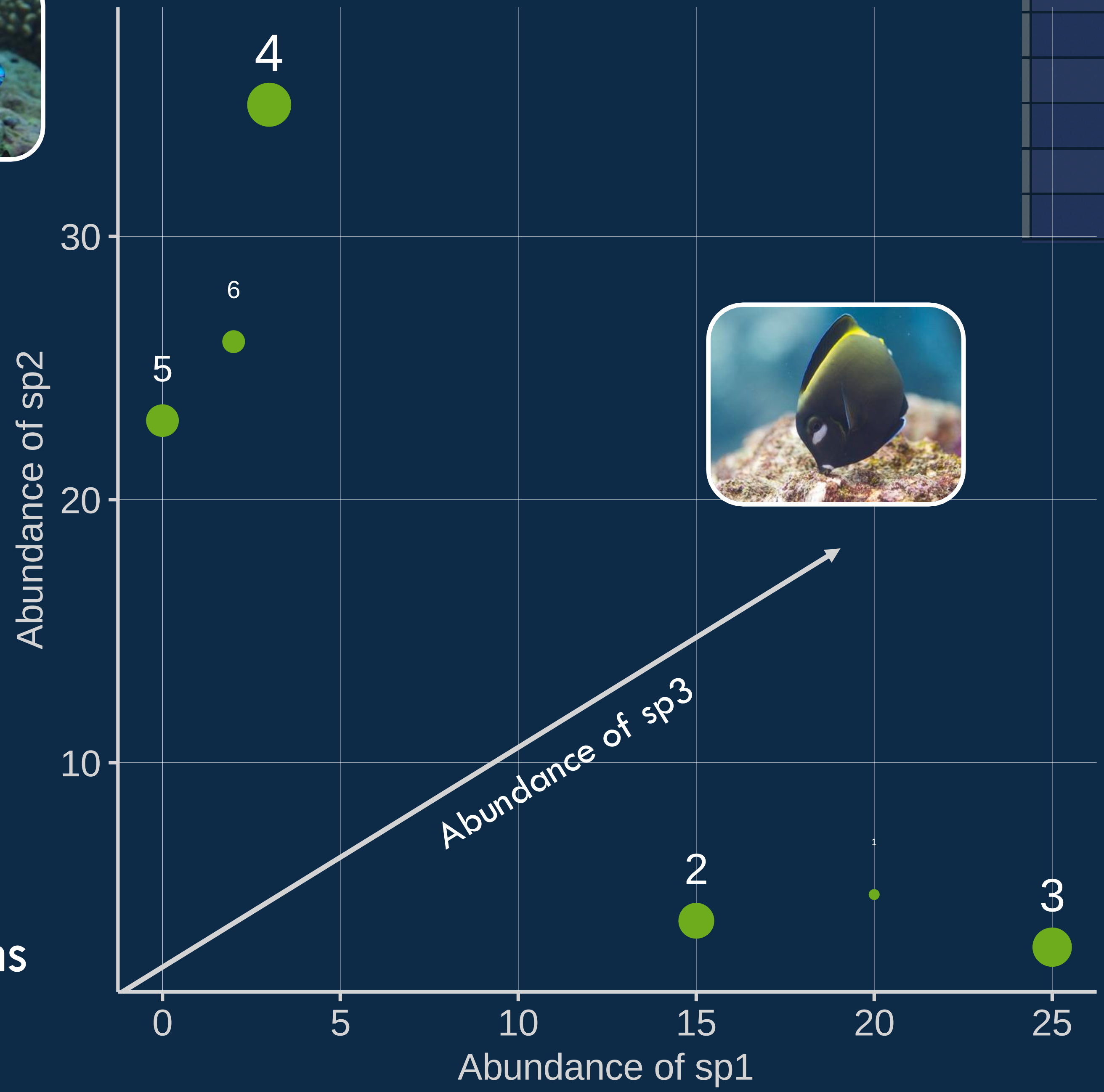
One dimension



Two dimensions



site	alin	cmic	ablo	sovi	anig
1	20	10	1	5	2
2	15	8	2	4	1
3	25	12	2	3	3
4	3	5	33	35	25
5	0	1	3	23	2
6	2	1	20	26	5



Three dimensions

FOUR STEPS OF PCA

1 Standardization: scaling and centering

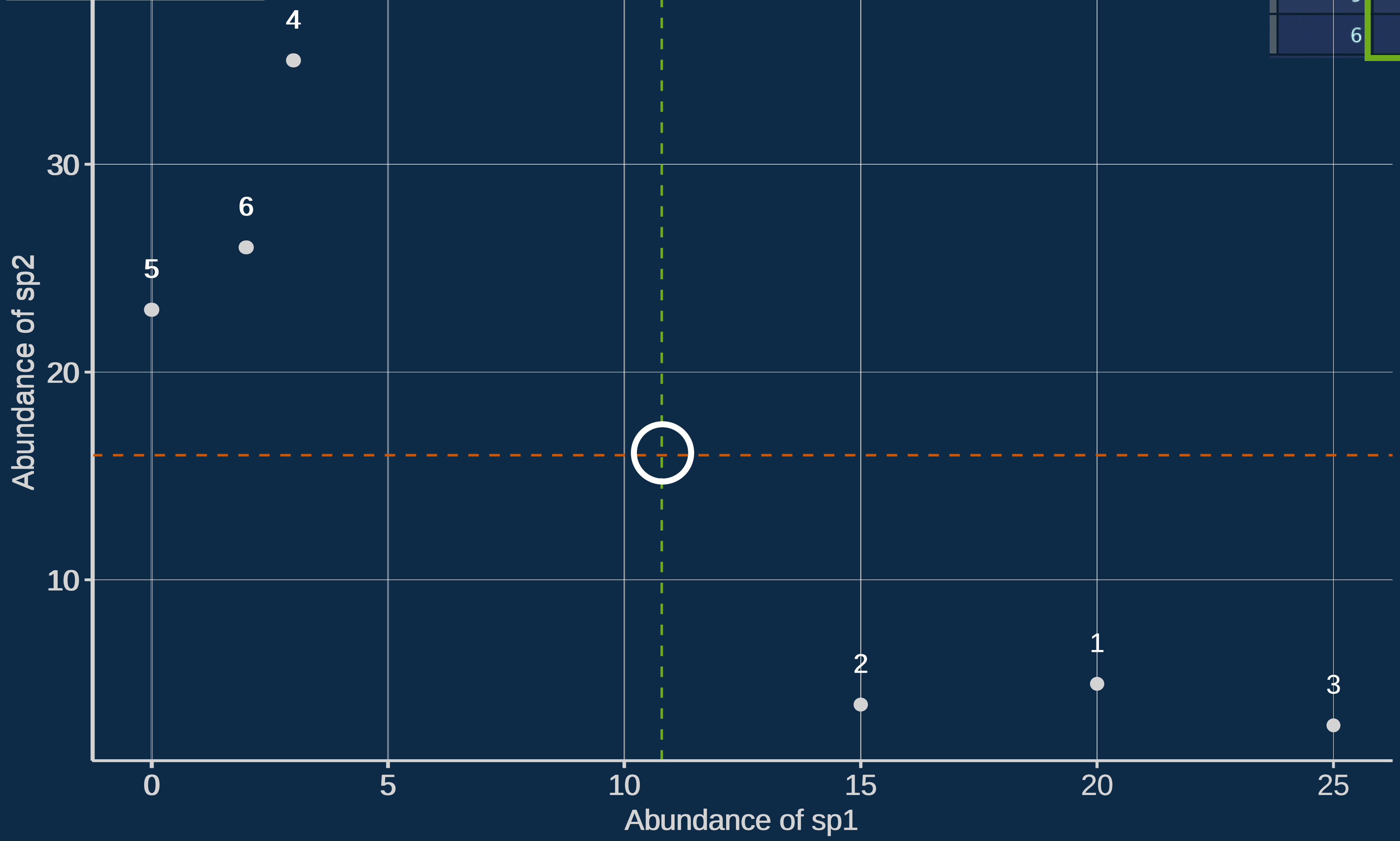
Compute a covariance matrix between variables

Compute eigenvectors and eigenvalues of the covariance matrix to identify principal components

Rotate data along the principal components axes

Principal Component Analysis

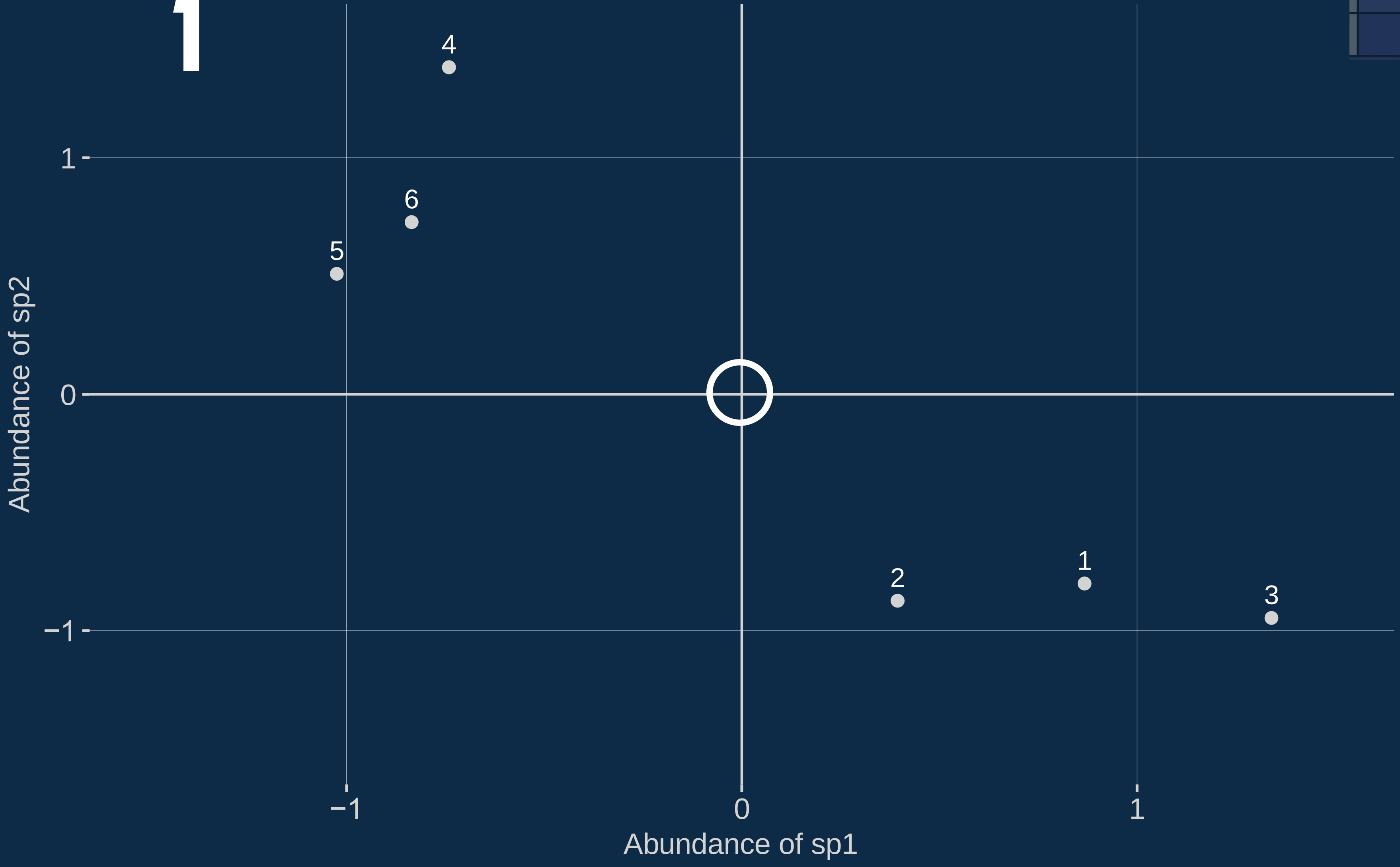
site	alin	cmic	ablo	sovi	anig
1	20	10	1	5	2
2	15	8	2	4	1
3	25	12	2	3	3
4	3	5	33	35	25
5	0	1	3	23	2
6	2	1	20	26	5



Standardized values

1

site	alin	cmic	ablo	sovi	anig
1	20	10	1	5	2
2	15	8	2	4	1
3	25	12	2	3	3
4	3	5	33	35	25
5	0	1	3	23	2
6	2	1	20	26	5



FOUR STEPS OF PCA

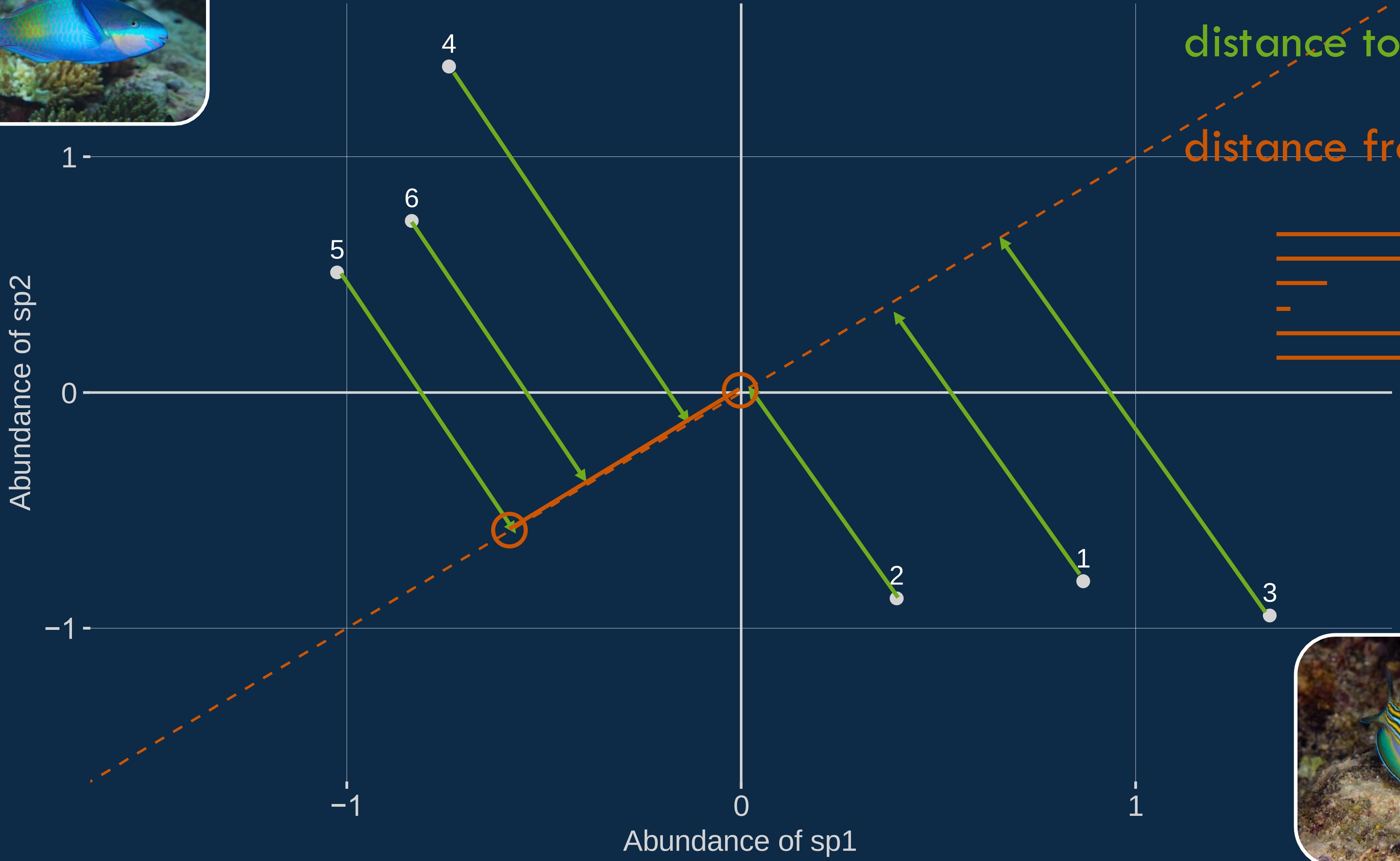
Standardization: scaling and centering

2 Compute a covariance matrix between variables

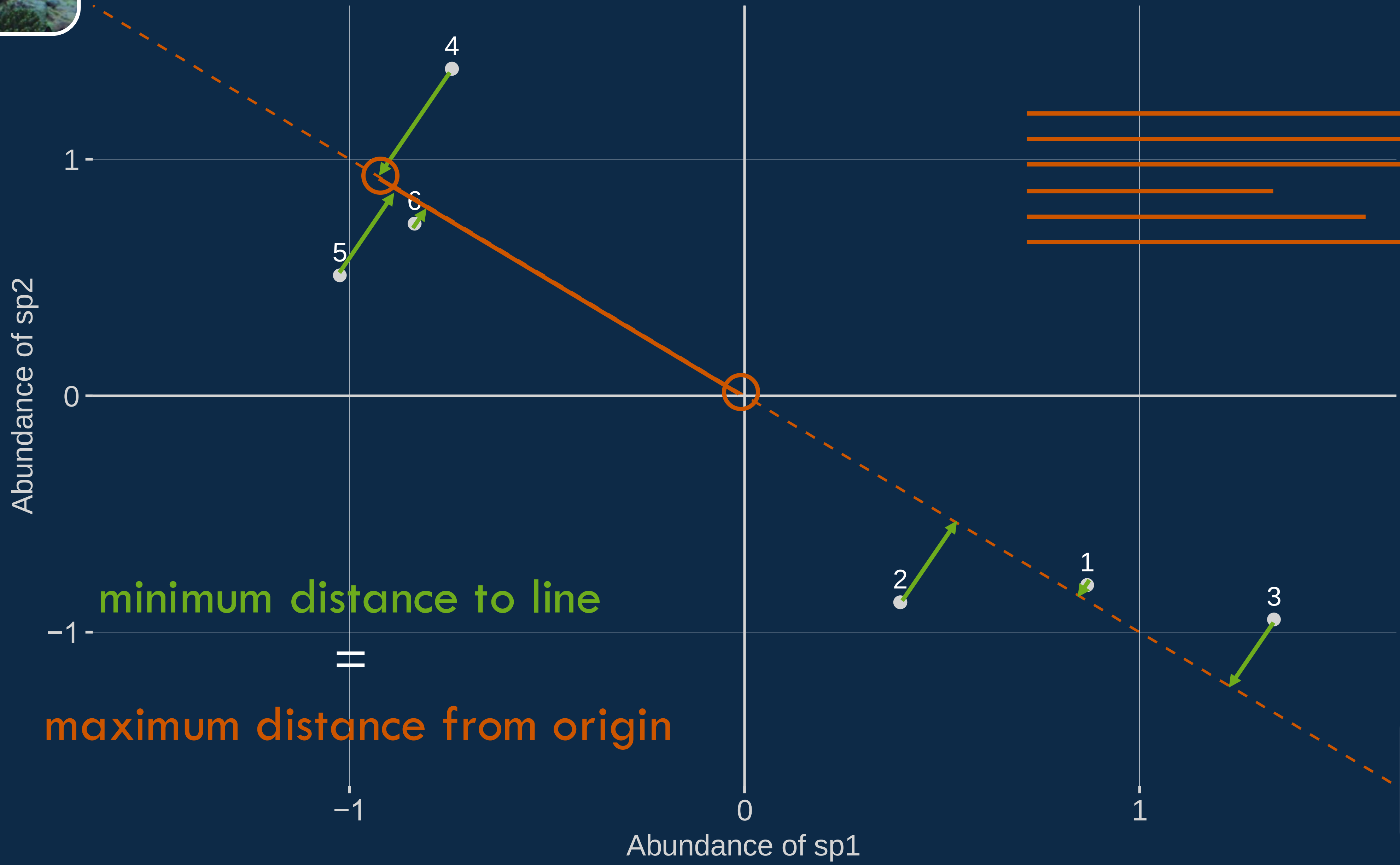
3 Compute eigenvectors and eigenvalues of the covariance matrix to identify principal components

Rotate data along the principal components axes

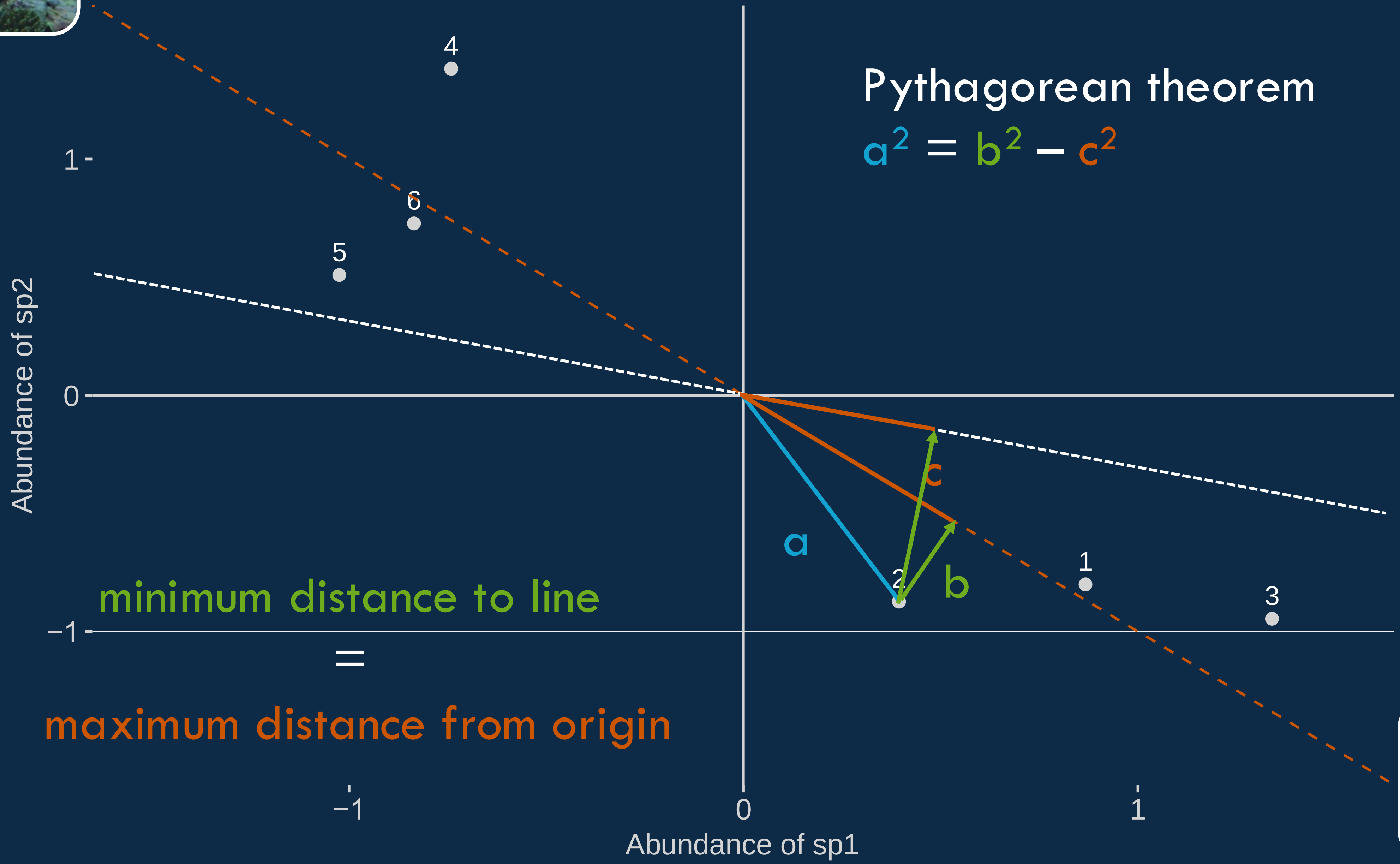
Line of best fit



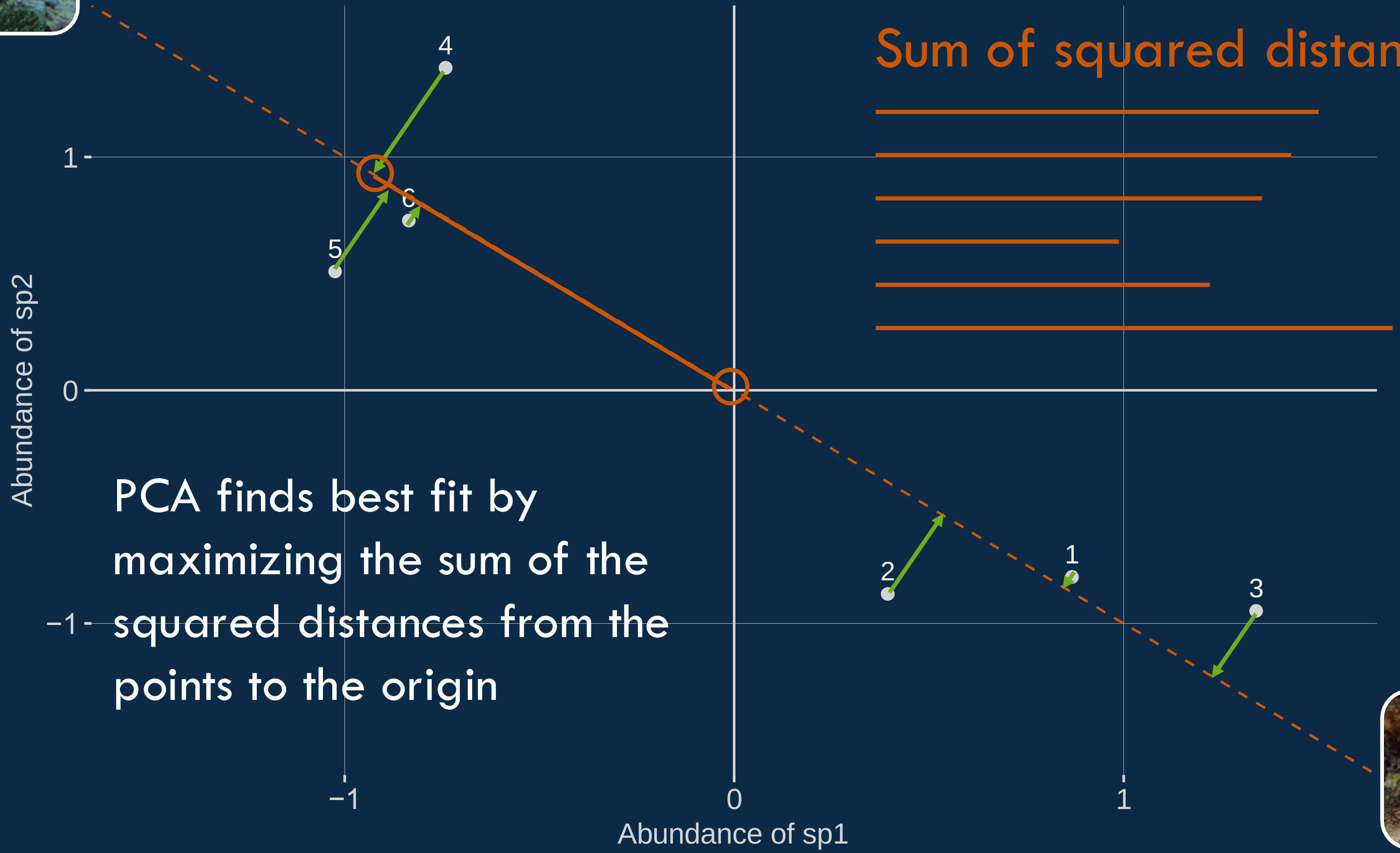
Line of best fit



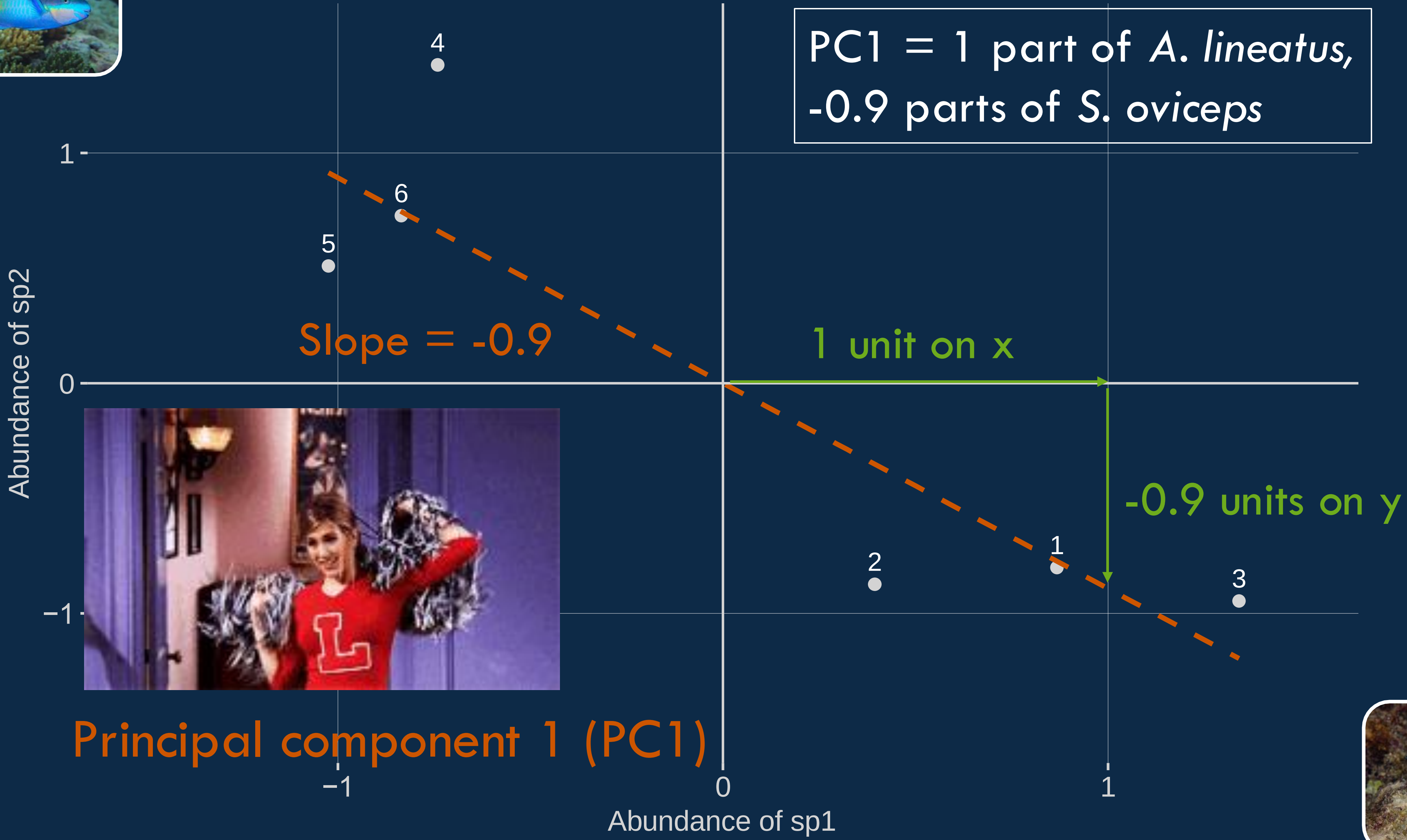
Line of best fit

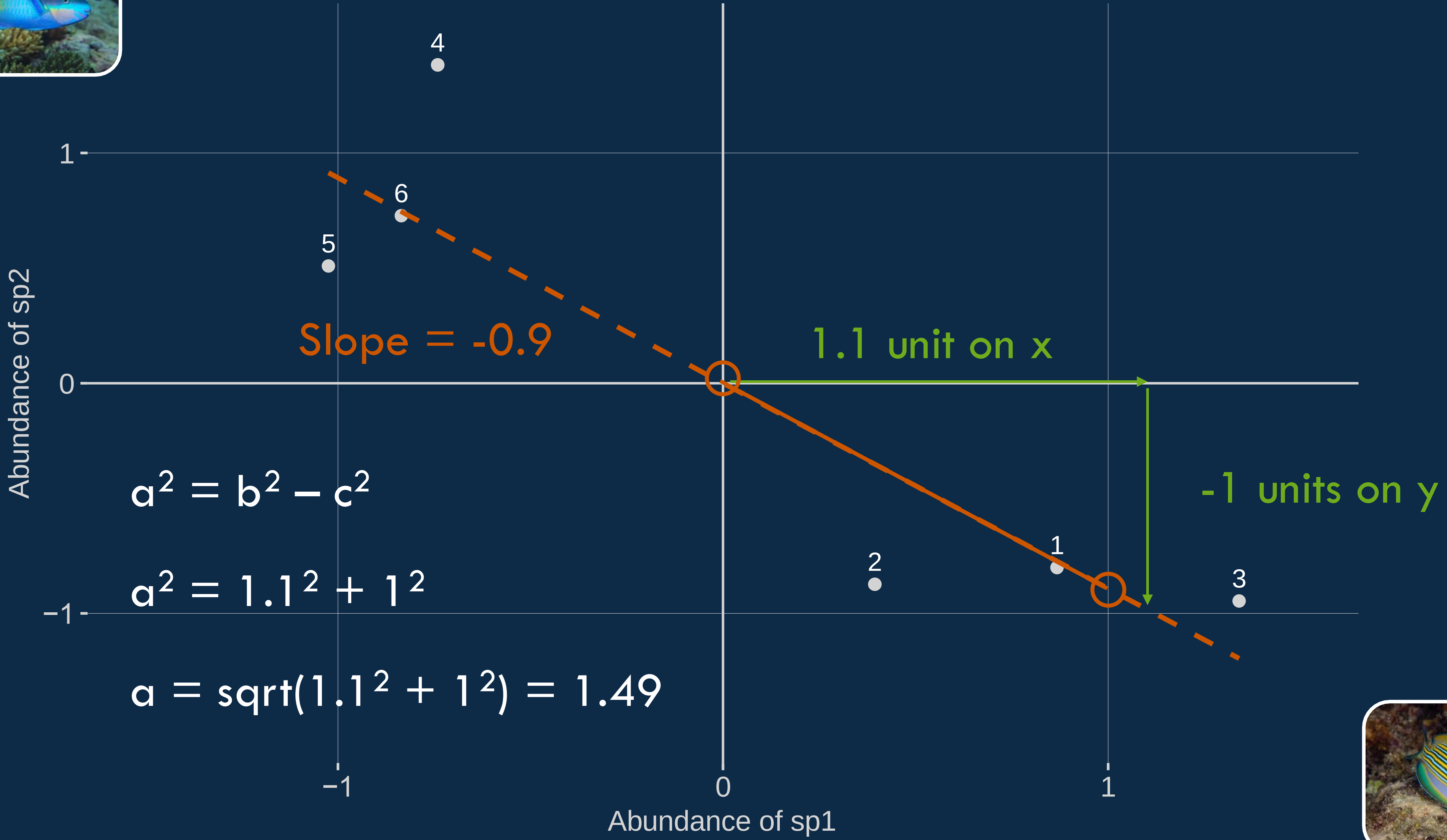


Line of best fit



PCA finds best fit by maximizing the sum of the squared distances from the points to the origin

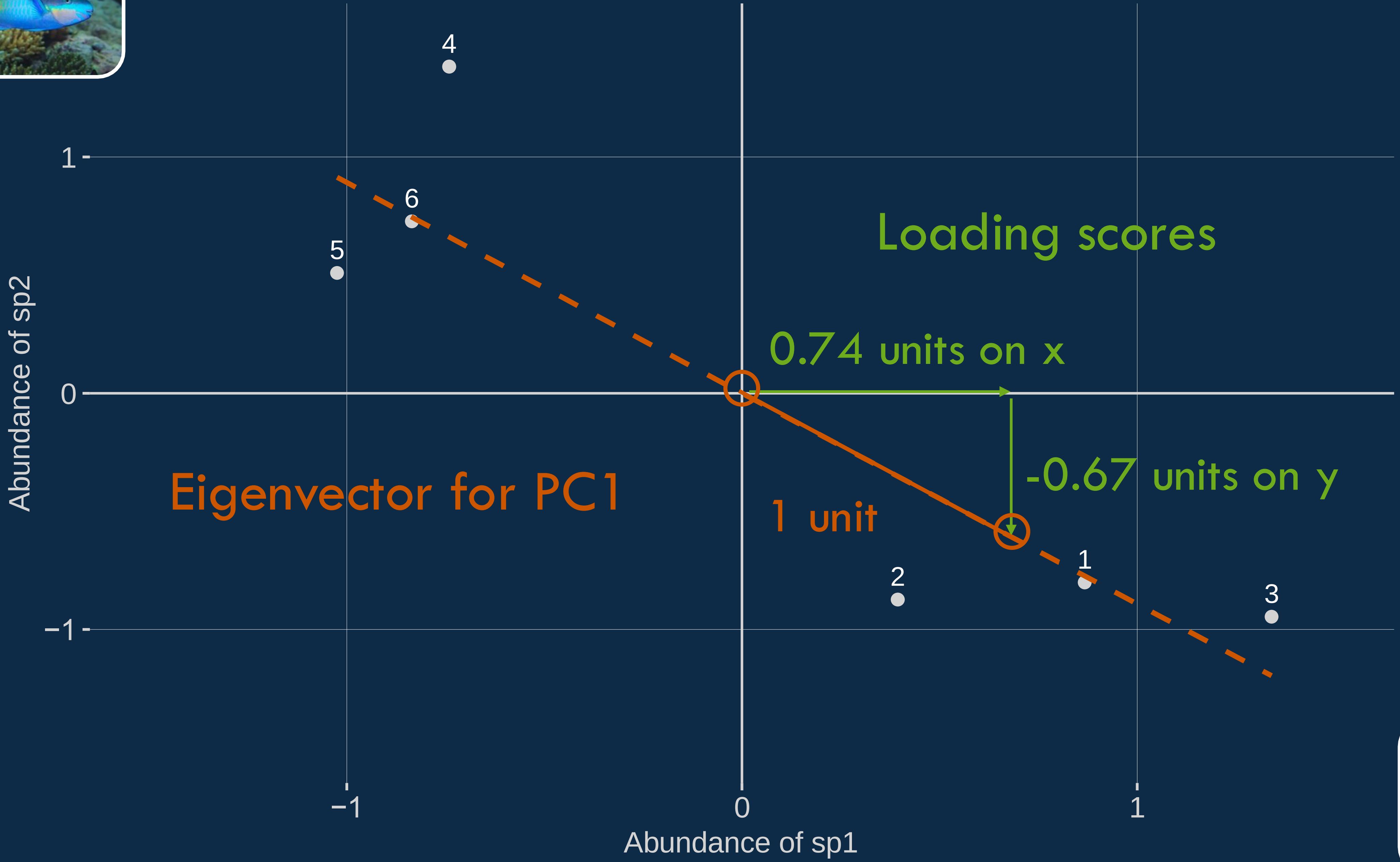

PC1 = 1 part of *A. lineatus*,
-0.9 parts of *S. oviceps*
Slope = -0.9
1 unit on x
-0.9 units on y
Principal component 1 (PC1)
Abundance of sp2
Abundance of sp1
1
0
−1
−1
0
1
4
6
5
2
1
3



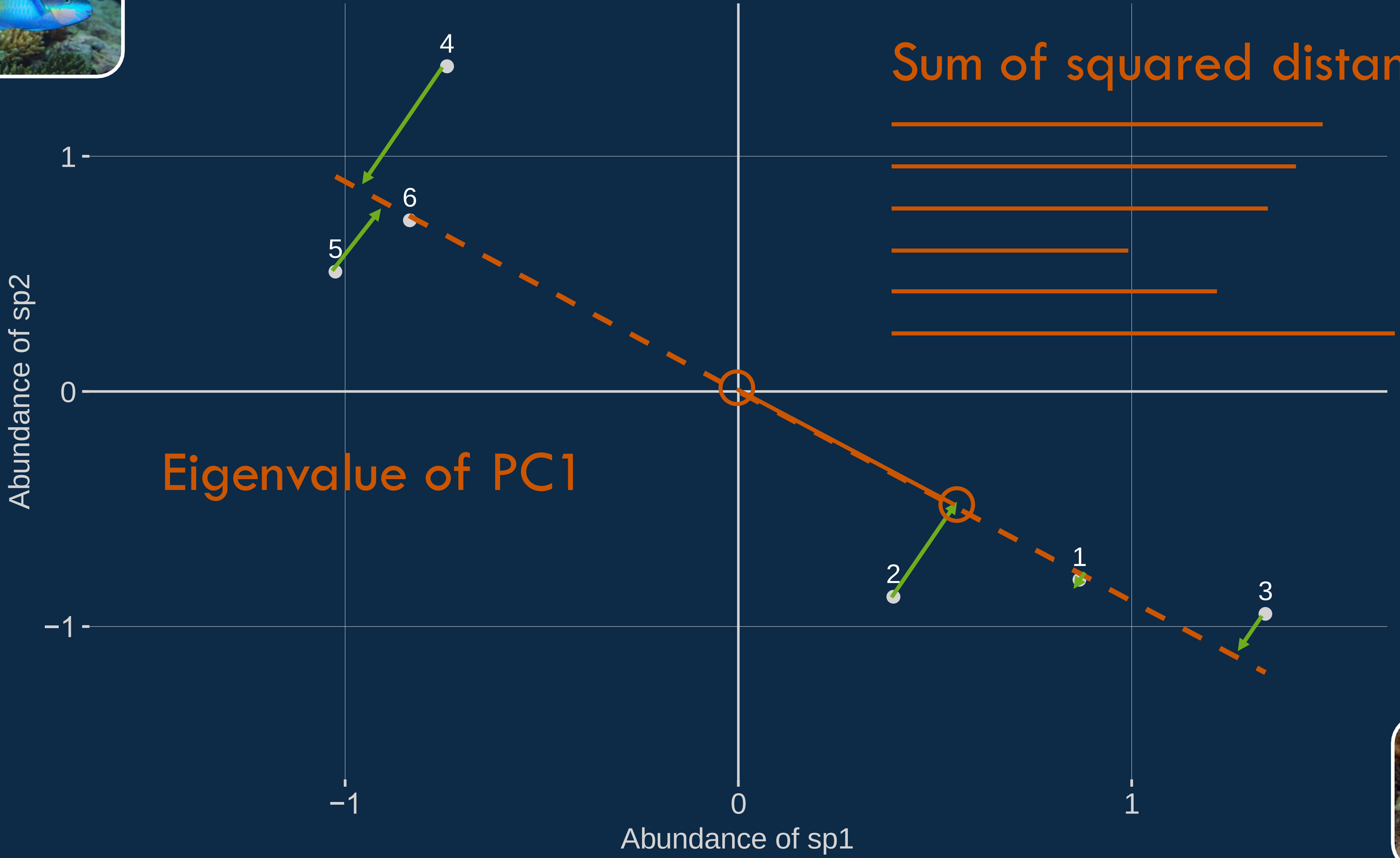
$$a^2 = b^2 - c^2$$

$$a^2 = 1.1^2 + 1^2$$

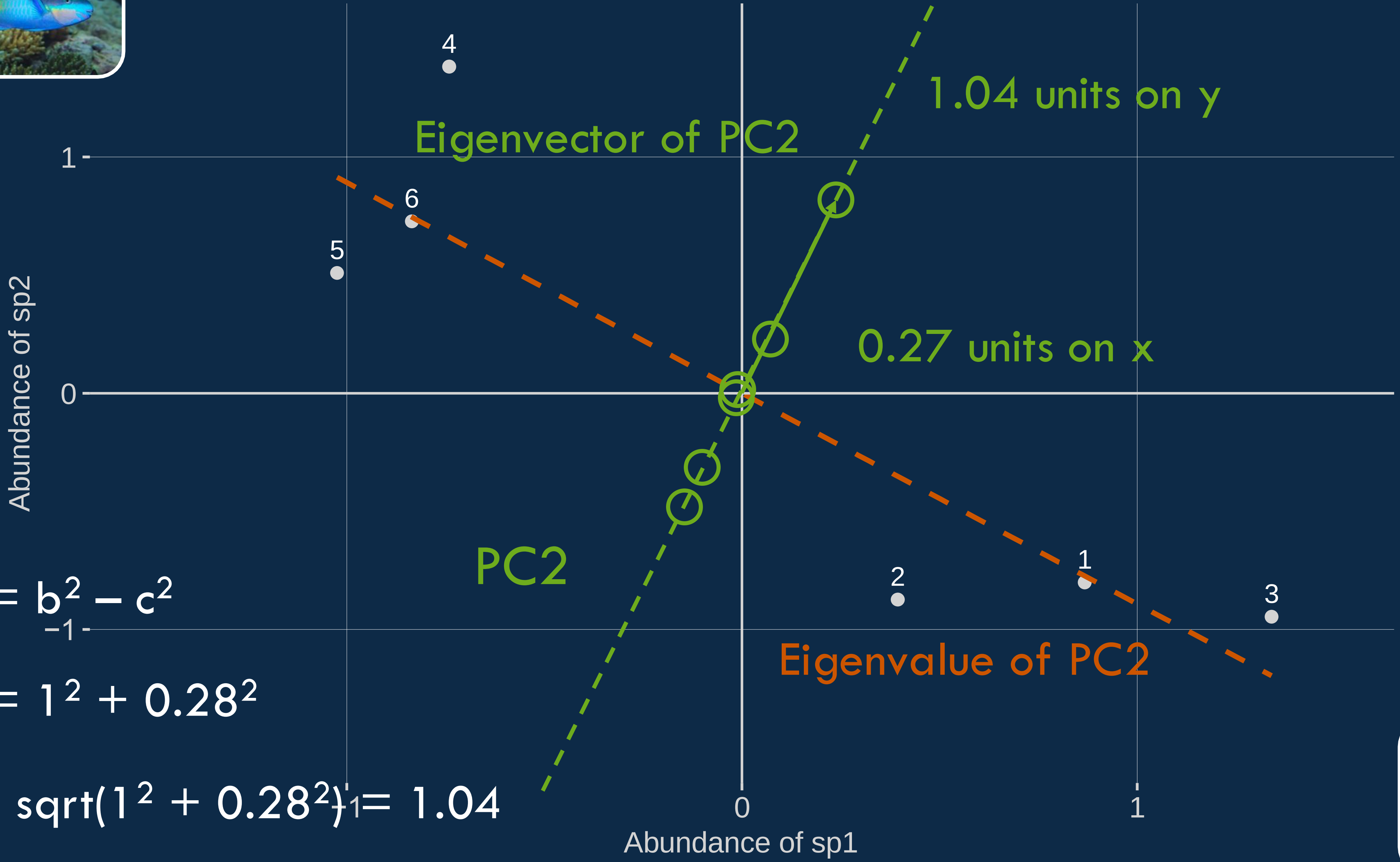
$$a = \text{sqrt}(1.1^2 + 1^2) = 1.49$$



Loading scores
0.74 units on x
-0.67 units on y
Eigenvector for PC1
1 unit
4
6
5
1
2
3
1
0
−1
−1
0
1
Abundance of sp2
Abundance of sp1


Sum of squared distances
Eigenvalue of PC1
4
6
5
2
1
3
1
0
−1
−1
0
1
Abundance of sp2
Abundance of sp1

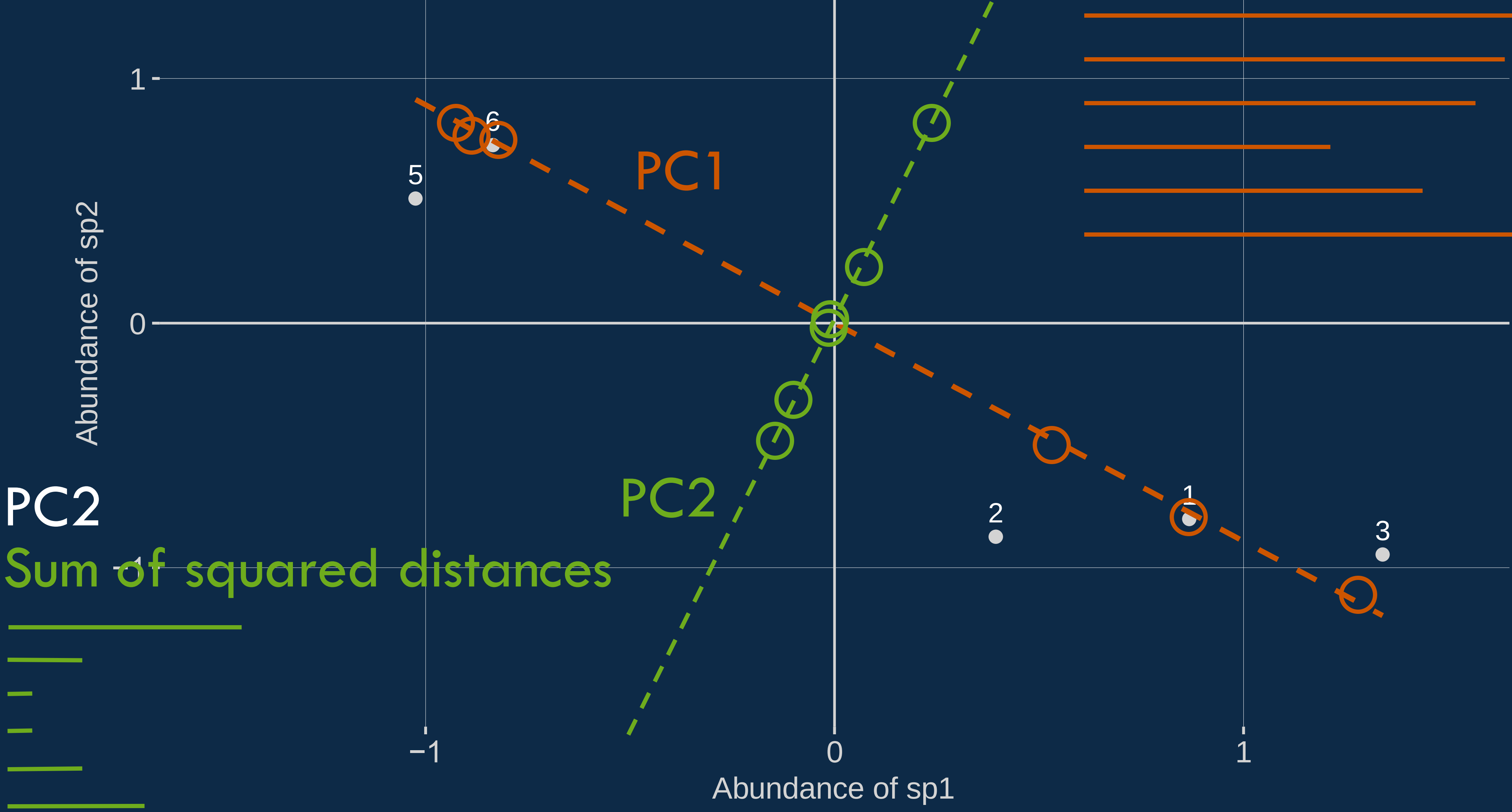
Species 2 (*S. oviceps*) is 4 times as important on PC2!



$a^2 = b^2 - c^2$

$a^2 = 1^2 + 0.28^2$

$a = \mathrm{sqrt}(1^2 + 0.28^2) = 1.04$


PC1
Sum of squared distances
PC2
Sum of squared distances
PC1
PC2
4
6
5
1
2
3
1
0
-1
−1
0
1
Abundance of sp1
Abundance of sp2

PC1

Sum of squared distances

PC2

Sum of squared distances

Eigenvalues/n-1 = % variance explained

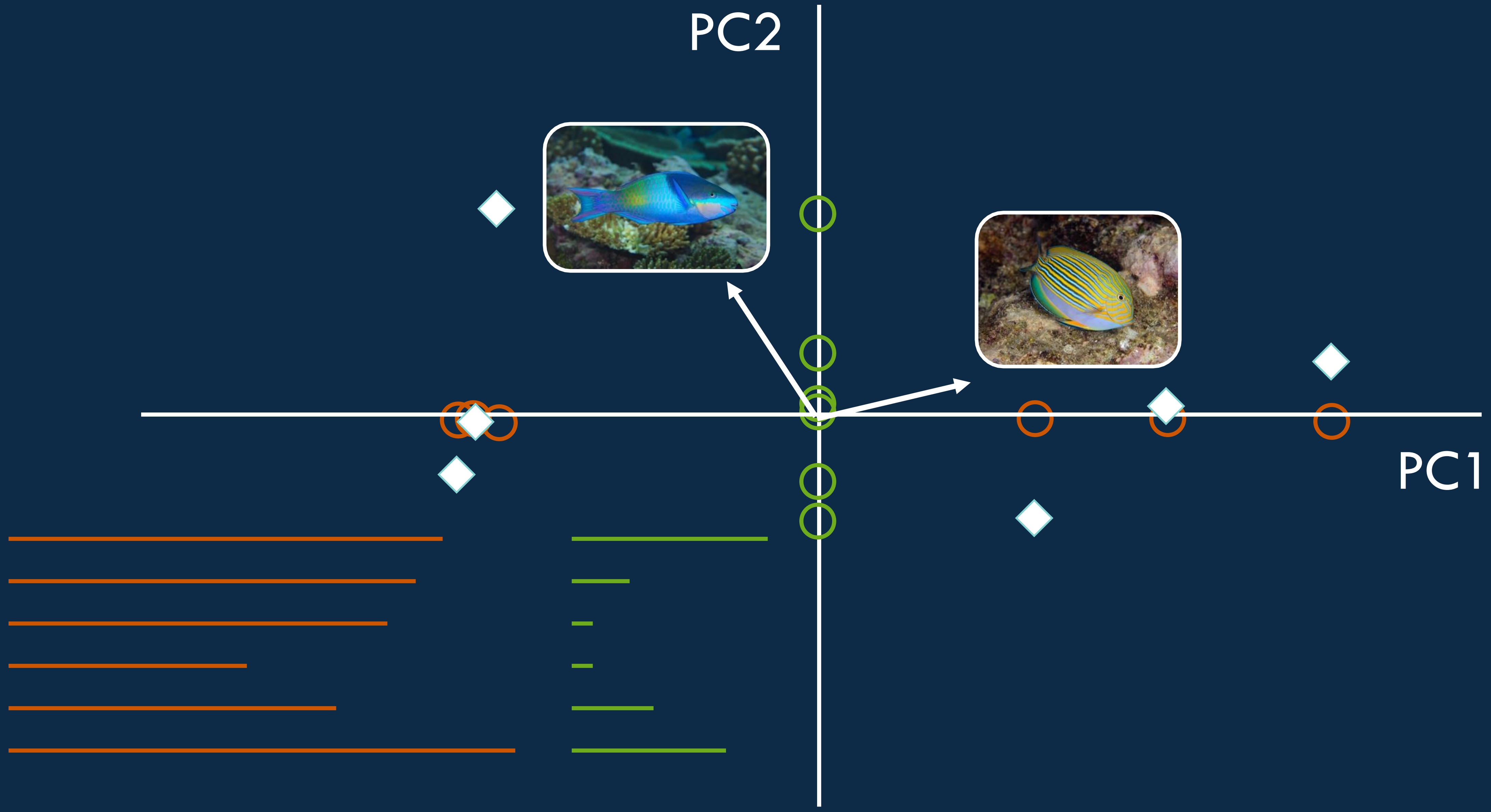
FOUR STEPS OF PCA

Standardization: scaling and centering

Compute a covariance matrix between variables

Compute eigenvectors and eigenvalues of the covariance matrix to identify principal components

4 Rotate data along the principal components axes



Eigenvalues/n-1 = % variance explained

FOUR STEPS OF PCA

Standardization: scaling and centering

Compute a covariance matrix between variables

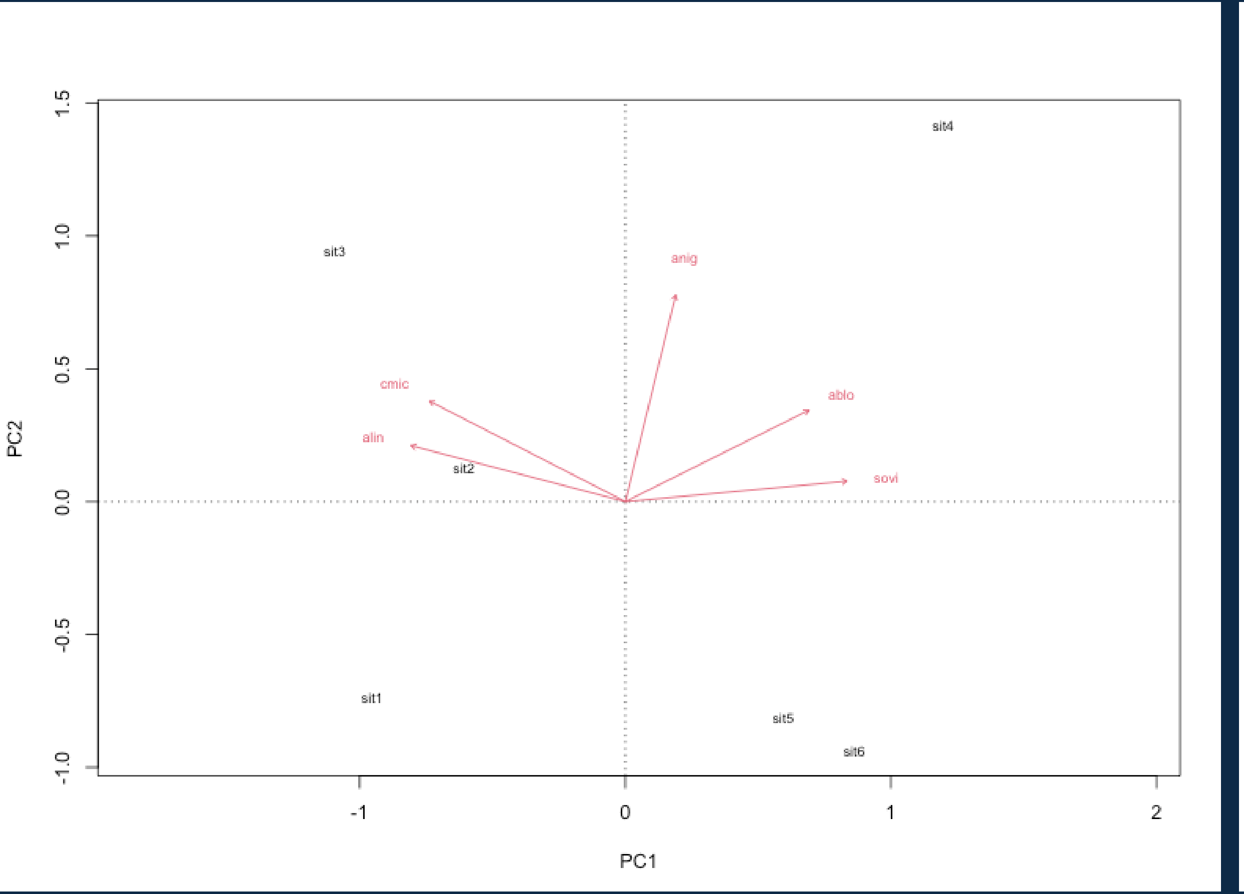
Compute eigenvectors and eigenvalues of the covariance matrix to identify principal components

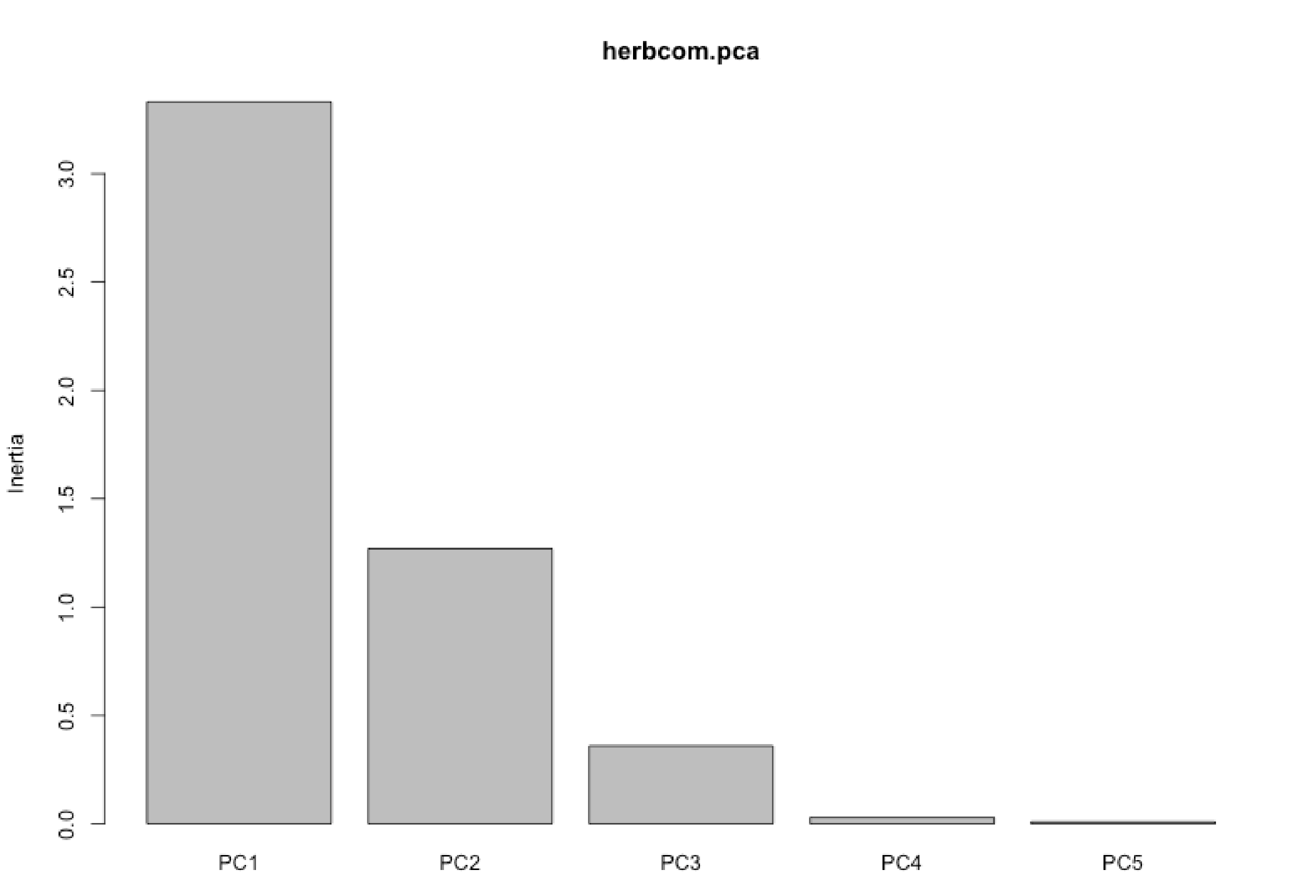
Rotate data along the principal components axes

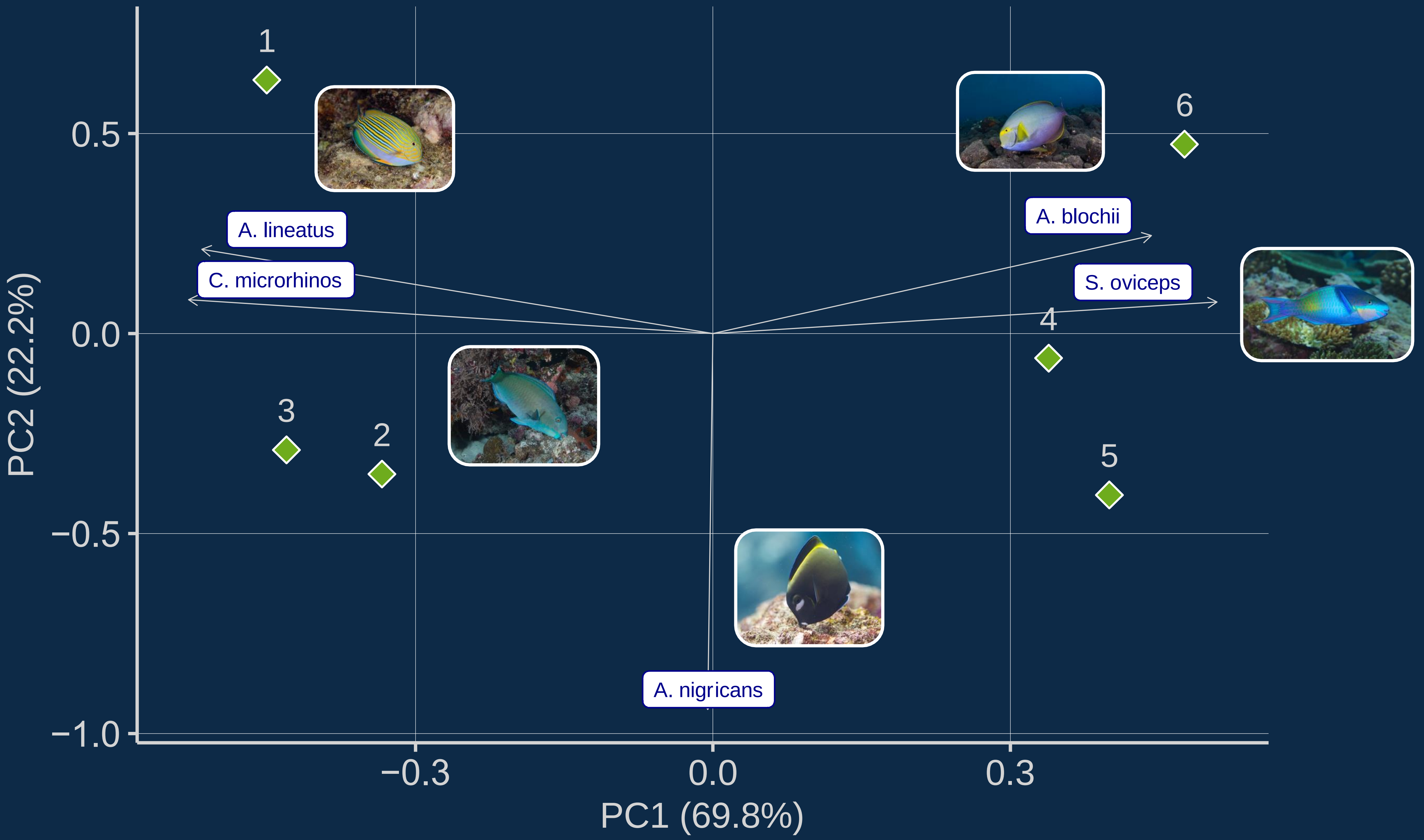
How many lines of code does it take to run a PCA in R?

How many lines of code does it take to run a PCA?

```
herb.com.pca <- rda(herb.com[-1], scale = T)
biplot(herb.com.pca)
```







1
2
3
4
5
6
A. lineatus
C. microrhinos
A. blochii
S. oviceps
A. nigricans
0.5
0.0
−0.5
−1.0
−0.3
0.0
0.3
PC2 (22.2%)
PC1 (69.8%)

BEYOND PCA

(because I actually shouldn't have done it the way I did)

CORRESPONDENCE ANALYSIS (CA)

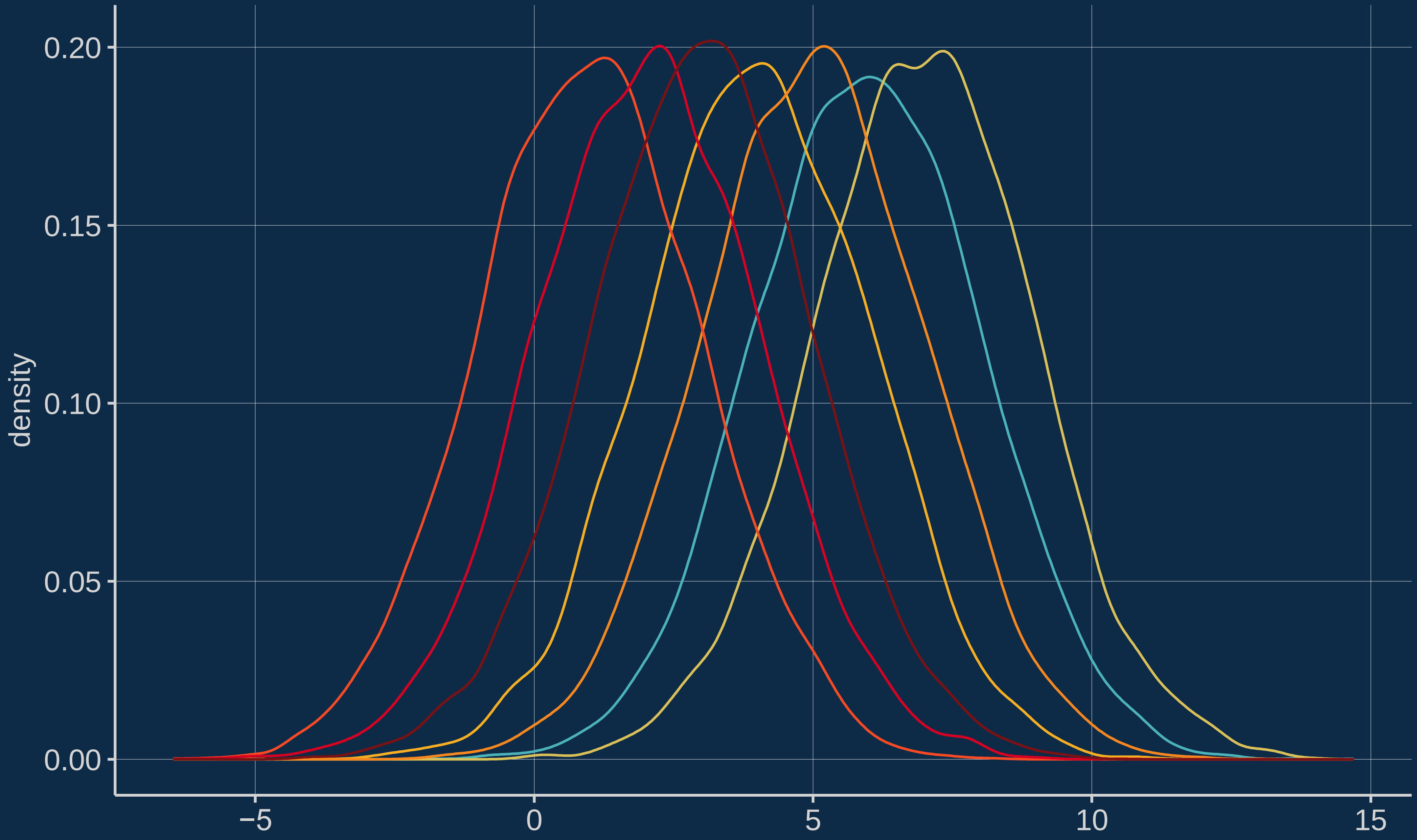
PRINCIPAL COORDINATES ANALYSIS (PCoA)

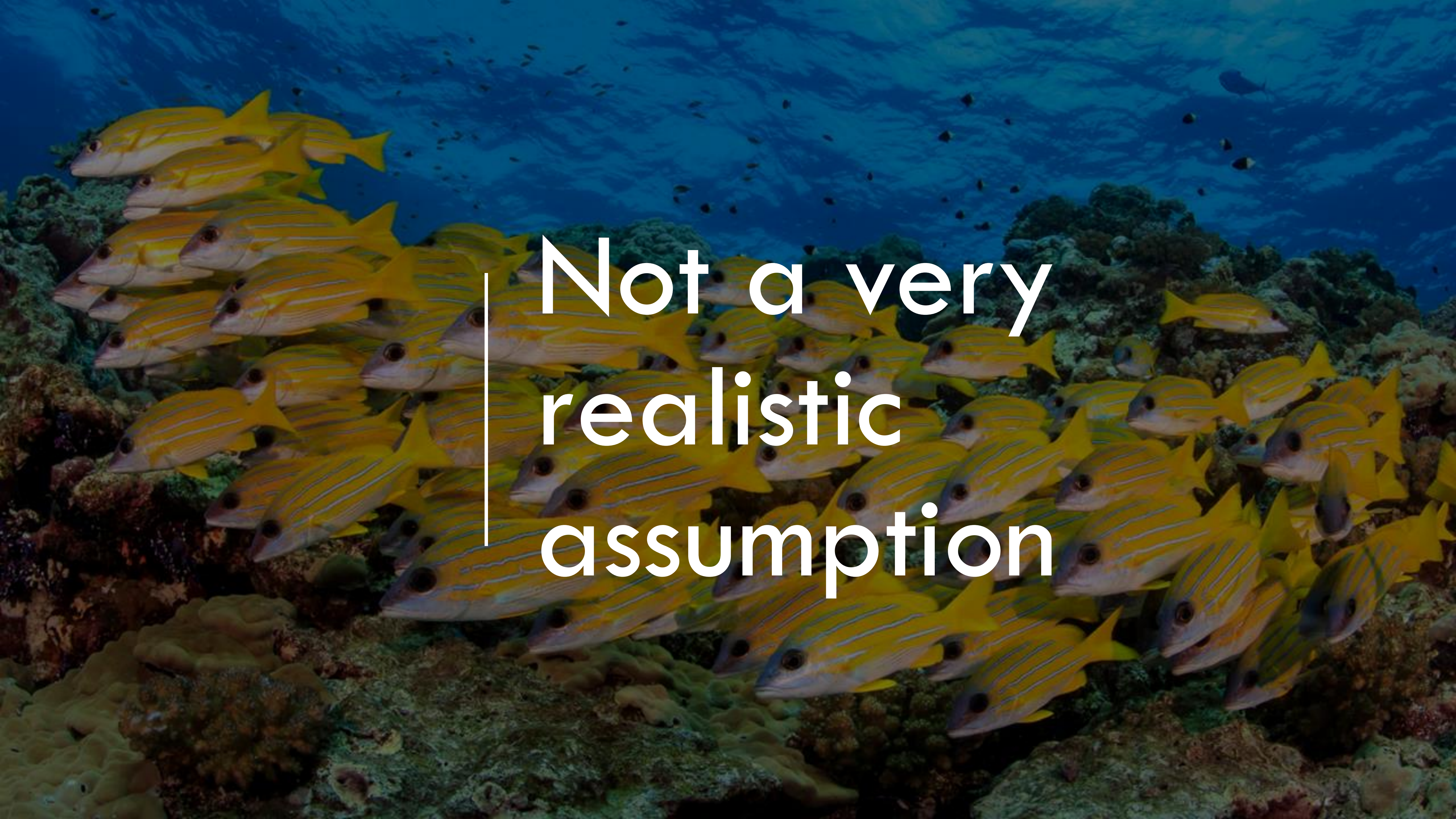
NON-METRIC MULTIDIMENSIONAL SCALING (nMDS)

CORRESPONDENCE ANALYSIS (CA)

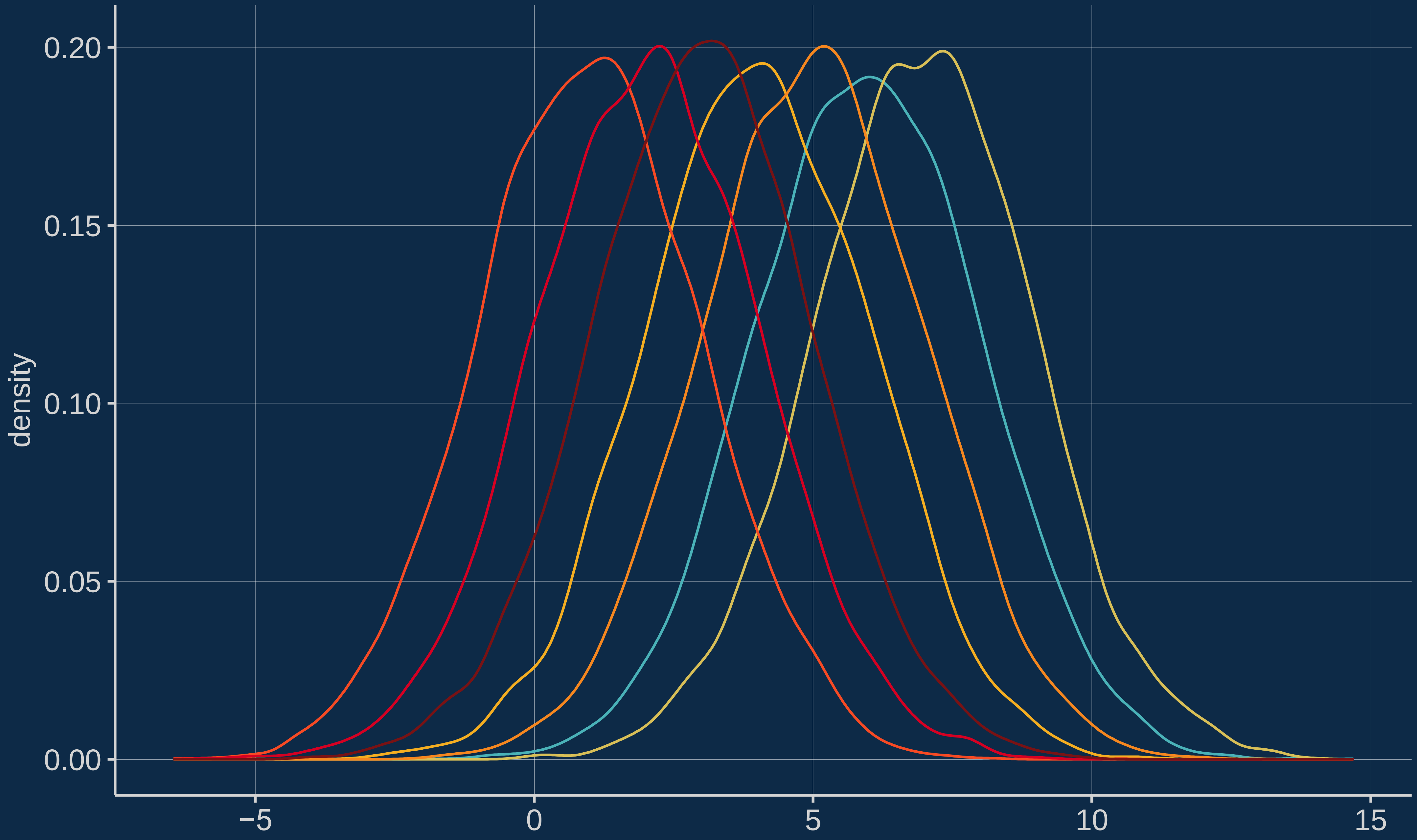
- Unimodal, unconstrained
- Basically a weighted PCA that works on relative rather than absolute abundances
- Each species distribution is a hump

Assumed to be unimodal Gaussian curves, with equal heights and variances



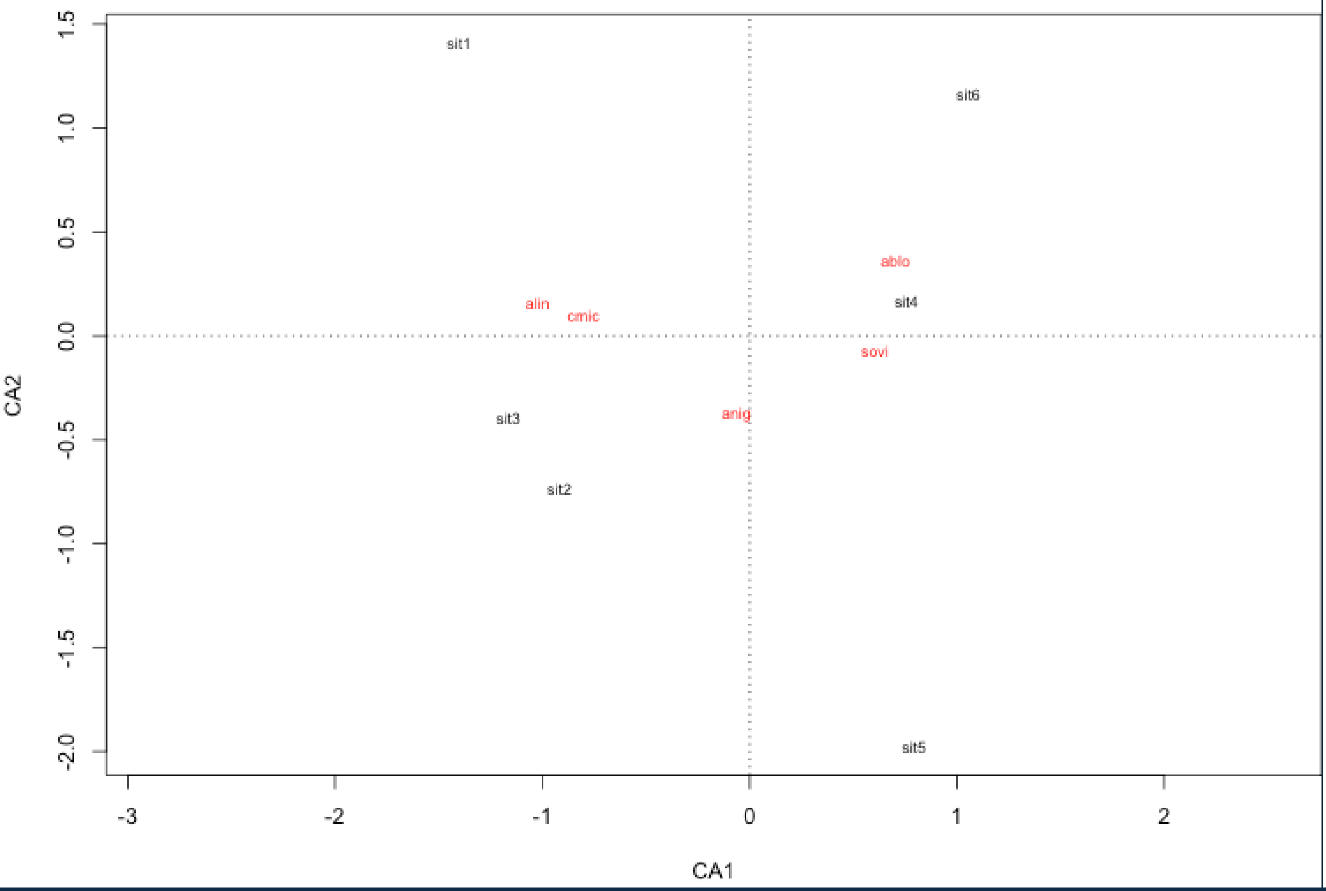

Not a very realistic assumption

CA is fairly robust to violations

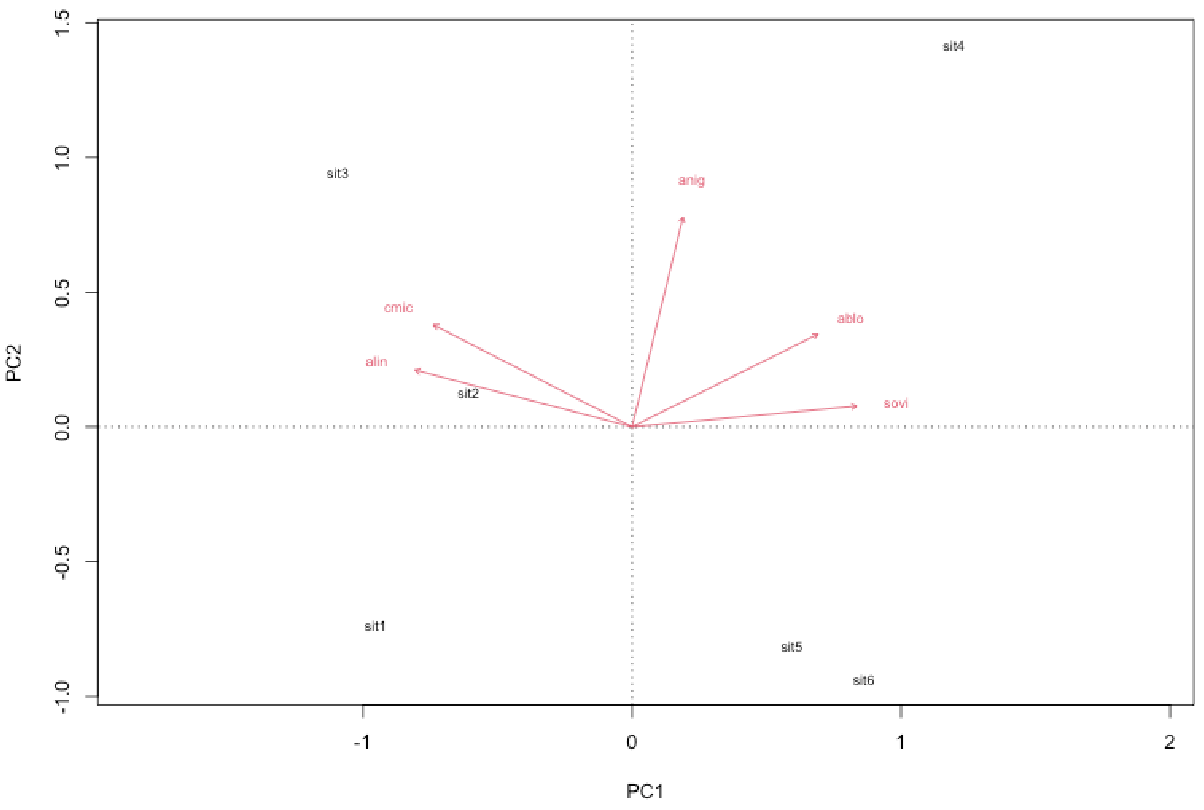


```
herb.cca <- cca(herb.com[-1])
```

CA



PCA



0s and 1s (presence/absence):
Use CA, not PCA

OR

Use a distance-based method

PRINCIPAL COORDINATES ANALYSIS (PCoA)

- distance-based ordination technique
- also known as metric multidimensional scaling (MDS)
- similar to PCA but based on distance matrices, not raw data
- PCoA with Euclidean distance = PCA

PRINCIPAL COORDINATES ANALYSIS (PCoA)

alin	cmic	ablo	sovi	anig
20	10	1	5	2
15	8	2	4	15
25	12	2	3	19
3	5	33	35	25
0	1	3	23	12
2	1	20	26	5

	1	2	3	4	5	6
1	0.0000000	0.2682927	0.2727273	0.7697842	0.7662338	0.7608696
2	0.2682927	0.0000000	0.1809524	0.6000000	0.5421687	0.7142857
3	0.2727273	0.1809524	0.0000000	0.6049383	0.6400000	0.7739130
4	0.7697842	0.6000000	0.6049383	0.0000000	0.4428571	0.3032258
5	0.7662338	0.5421687	0.6400000	0.4428571	0.0000000	0.3118280
6	0.7608696	0.7142857	0.7739130	0.3032258	0.3118280	0.0000000

Principal Coordinates Analysis (PCoA)

Distance/dissimilarity/similarity metrics to calculate dissimilarity matrices for PCoA

Euclidean, Bray-Curtis, Manhattan, Kulczynski, Jaccard, Gower, Mahalanobis, Canberra, Chi-squared, Canberra, Sorensen-Dice, Steinhaus, Ochai, Aitchinson, and many more…

The double zero problem

Indicate closeness of objects
Not metric unless converted to distance

Inverse of similarity measures
metric = distance
non-metric = dissimilarity

Similarity measures

- Binary measures (0s and 1s): presence or absence
  - Jaccard or Sørensen coefficient
  - Jaccard excludes 0-0, equally weights all else
  - Sørensen gives double weight to 1-1

- Quantitative measures: values other than 0 and 1
  - Gower's coefficient
  - Allows mixing and matching of data (binary, categorical, numeric, even NAs)

Distance and dissimilarity measures

Metric distances

- Euclidean - uses Pythagorean, no good for 0s
- Chord - relative proportions, insensitive to 0-0
- Mahalanobis - good for groups
- Hellinger - good for linear, rare species have low weights
- Canberra - excludes double 0s, rare species have high weights

Semi-metric measures

- Bray-Curtis - often used for raw count data, treats differences between high and low variable values equally

Non-metric measures

- AVOID!


Euclidean
Mahalanobis
Bray−Curtis
Dim1
Dim2
10
5
0
−5
−10
−20
−10
0
0.5
1.0
0.2
0.1
0.0
−0.1
−0.2
−0.4
−0.2
0.0
0.2
0.4
−0.2
0.0
0.2
0.4
1
2
3
4
5
6
WHAT IF I TOLD YOU
I DON'T CARE?

PCoA output

```
Call: wcmdscale(d = d, eig = TRUE)

           Inertia Rank
Total      0.81028
Real       0.86113    4
Imaginary -0.05085    1

Results have 6 points, 4 axes

Eigenvalues:
[1]  0.6417  0.1072  0.1061  0.0061 -0.0509

Weights: Constant
```

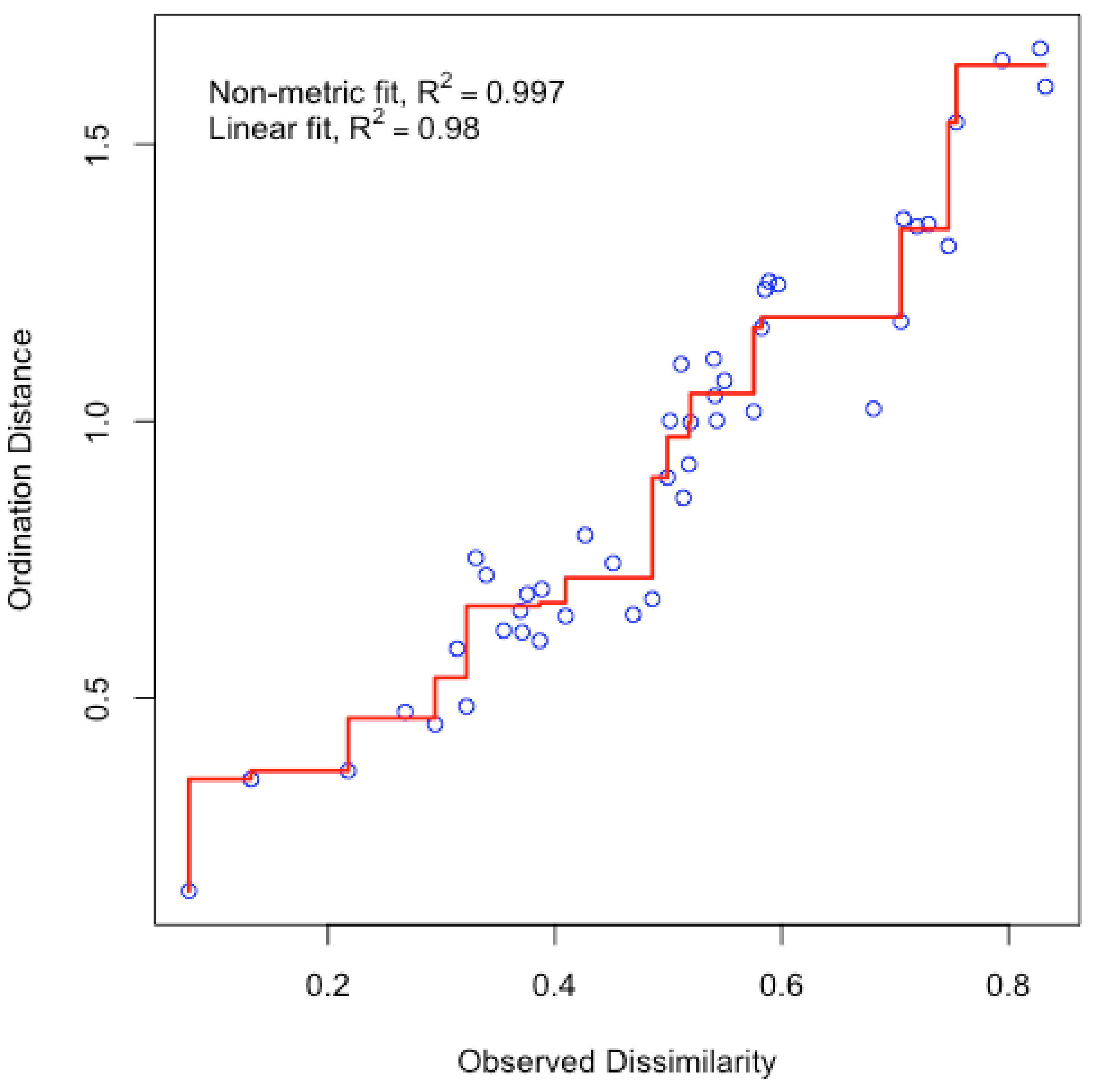
Negative eigenvalues can be corrected using either Lingoes or Cailliez transformation

NON-METRIC MULTIDIMENSIONAL SCALING (nMDS)

- similar to PCoA but non-metric
- works based on the rank order of dissimilarities

nMDS can't handle stress

- tries to map dissimilarity with lowest stress
- stress is the sum of squared residuals of regressions between distances
- iterative solution
- may not converge


Non-metric fit, $R^2 = 0.997$
Linear fit, $R^2 = 0.98$
Ordination Distance
0.5
1.0
1.5
0.2
0.4
0.6
0.8
Observed Dissimilarity

nMDS is very flexible

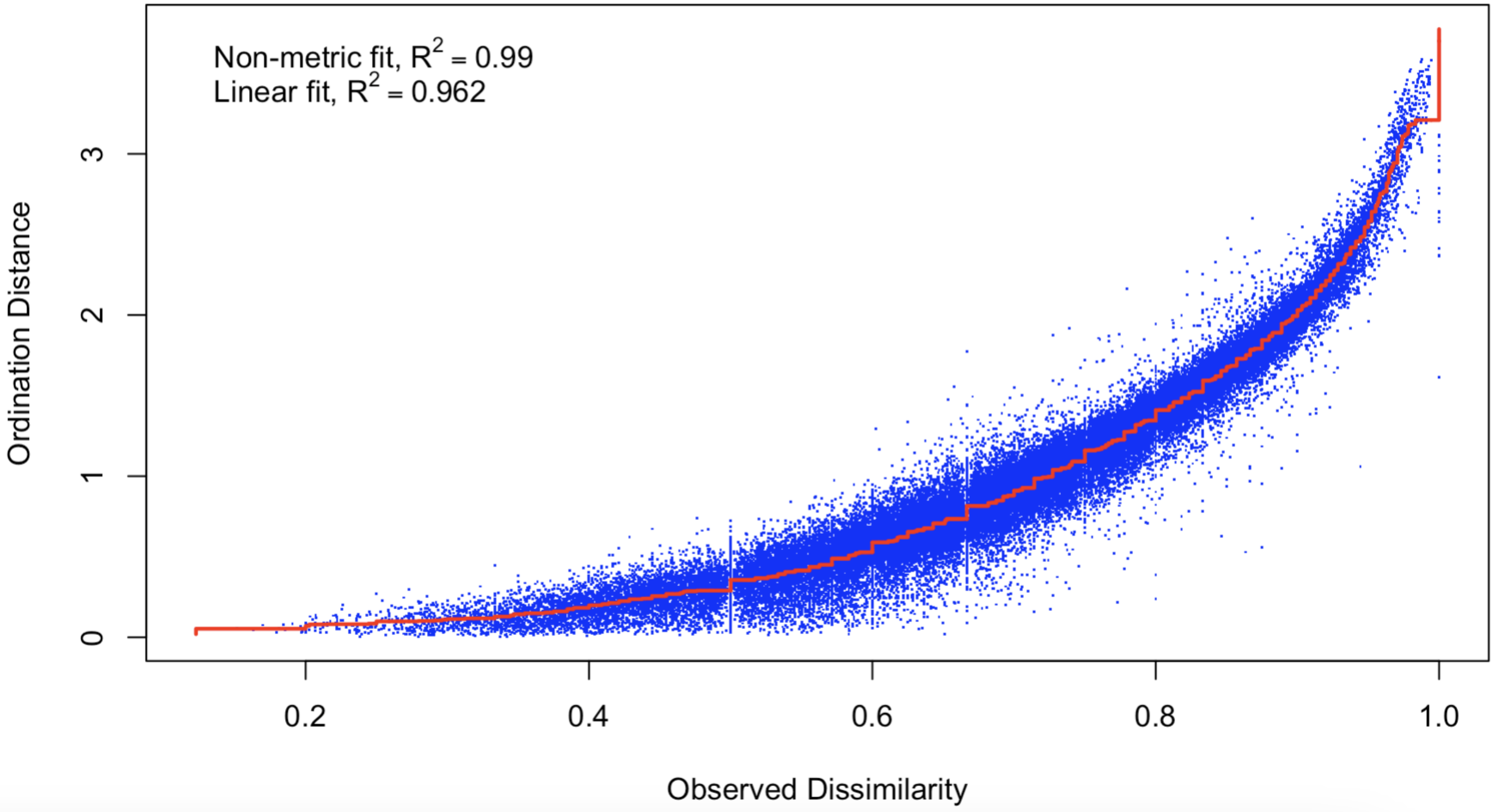
- nMDS is most flexible ordination technique
- low sample sizes are problematic: no stress, but also no results ☹
- example: historic bird data

Bird community data from 2010 to 2023

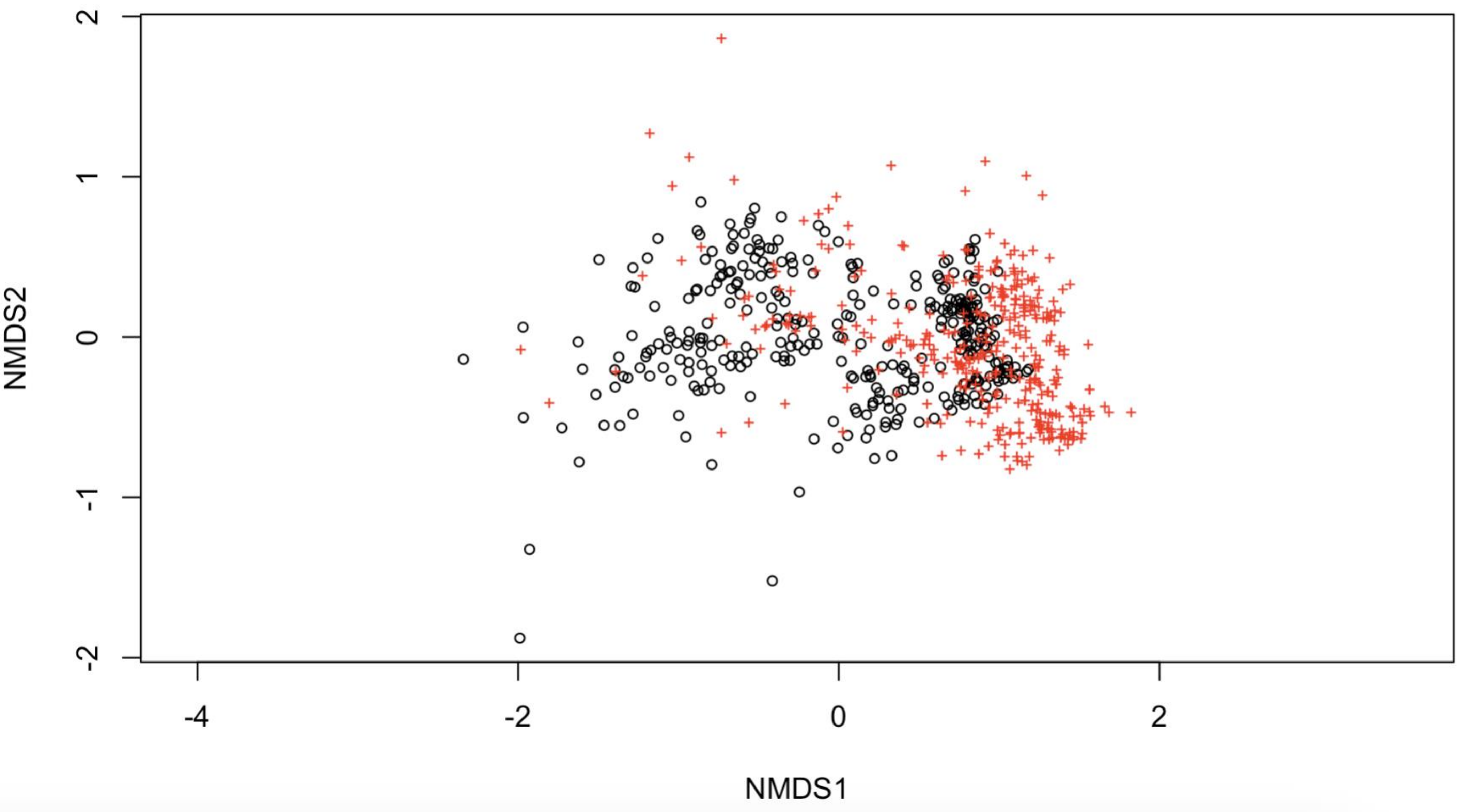
	year	month	location	Acadian Flycatcher	Alder Flycatcher	American Avocet	American Bittern	American Coot	American Flamingo
1	2010	Apr	Jetty	0	0	1	0	0	0
2	2010	Apr	Leonabelle	0	0	1	0	1	0
3	2010	Aug	Jetty	0	0	0	0	0	0
4	2010	Aug	Leonabelle	0	0	1	0	1	0
5	2010	Dec	Jetty	0	0	0	0	0	0
6	2010	Dec	Leonabelle	0	0	1	0	1	0
7	2010	Feb	Jetty	0	0	1	0	0	0
8	2010	Feb	Leonabelle	0	0	1	1	1	0
9	2010	Jan	Leonabelle	0	0	1	1	1	0
10	2010	Jul	Jetty	0	0	0	0	0	0
11	2010	Jul	Leonabelle	0	0	1	0	0	0
12	2010	Jun	Jetty	0	0	0	0	0	0
13	2010	Jun	Leonabelle	0	0	0	0	1	0
14	2010	Mar	Jetty	0	0	1	0	0	0
15	2010	Mar	Leonabelle	0	0	1	1	1	0
16	2010	May	Leonabelle	0	0	1	0	1	0
17	2010	Nov	Jetty	0	0	0	0	0	0
18	2010	Nov	Leonabelle	0	0	1	0	1	0

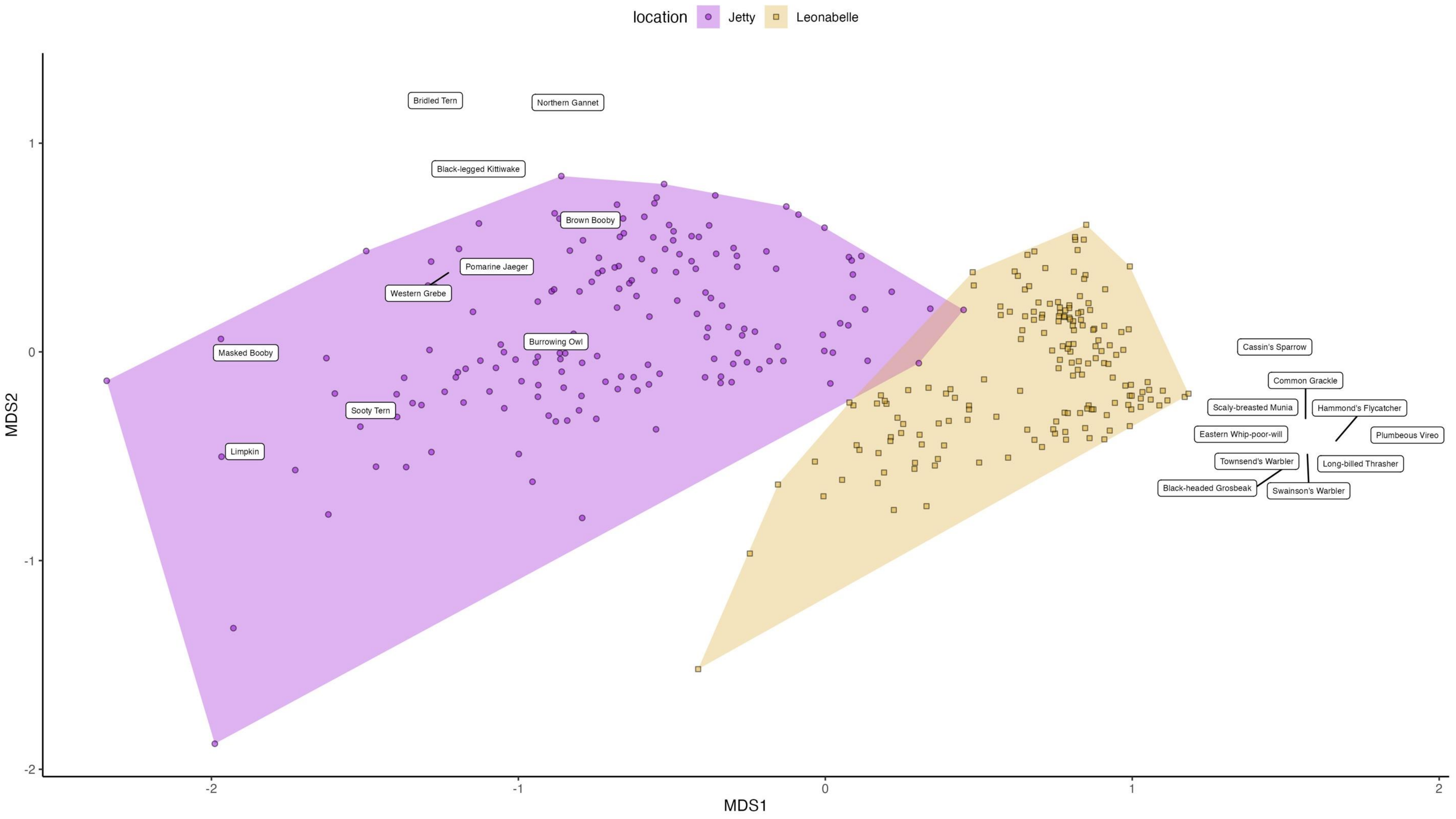
nMDS on Jaccard distance

```
bird.data.mds <- metaMDS(
bird.data.wide[-c(1:3)],
distance = "jaccard",
binary = TRUE,
trymax = 100)
```



Non-metric fit, R^2 = 0.99
Linear fit, R^2 = 0.962
Ordination Distance
0
1
2
3
0.2
0.4
0.6
0.8
1.0
Observed Dissimilarity


2
1
0
-1
-2
NMDS2
-4
-2
0
2
NMDS1



location
Jetty
Leonabelle
MDS2
MDS1
1
0
-1
-2
-2
-1
0
1
2
Bridled Tern
Northern Gannet
Black-legged Kittiwake
Brown Booby
Pomarine Jaeger
Western Grebe
Burrowing Owl
Masked Booby
Sooty Tern
Limpkin
Cassin's Sparrow
Common Grackle
Scaly-breasted Munia
Hammond's Flycatcher
Eastern Whip-poor-will
Plumbeous Vireo
Townsend's Warbler
Long-billed Thrasher
Black-headed Grosbeak
Swainson's Warbler

Redundancy analysis (RDA)

- constrained form of PCA

- allows quantification of how much variance an explanatory variable explains (response variable that are "redundant") with the explanatory variables

- functions through multiple regression analyses and PCA

Canonical Correspondence Analysis (CCA)

- constrained form of Correspondence Analysis
- works with non-linear data (including 0s and 1s)
- practically similar to RDA


location
a Jetty
a Leonabelle
Western Gull
Spotted Towhee
Trindade Petrel
Black-legged Kittiwake
Elegant Tern
Bridled Tern
Brown Noddy
Sabine's Gull
locationJetty
Burrowing Owl
Limpkin
Philadelphia Vireo
Acadian Flycatcher
Green Kingfisher
Winter Wren
American Bittern
Ross's Goose
Black-and-white Warbler
locationLeonabelle
Northern Bobwhite
Greater Flamingo
CA2
CCA1
2
0
-2
-4
-1
0
1
2
3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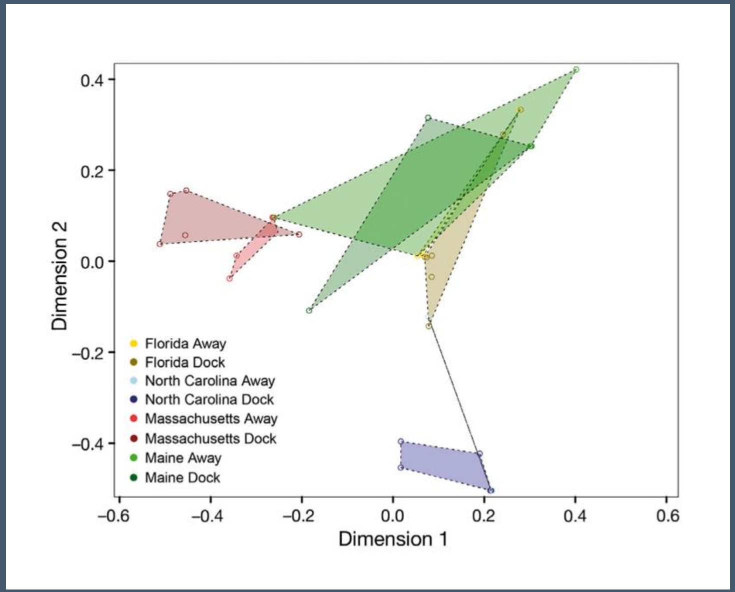
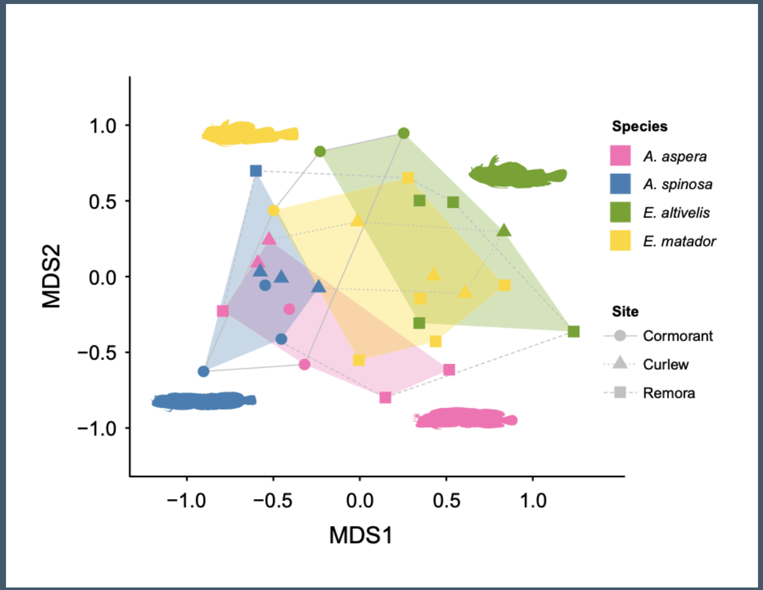
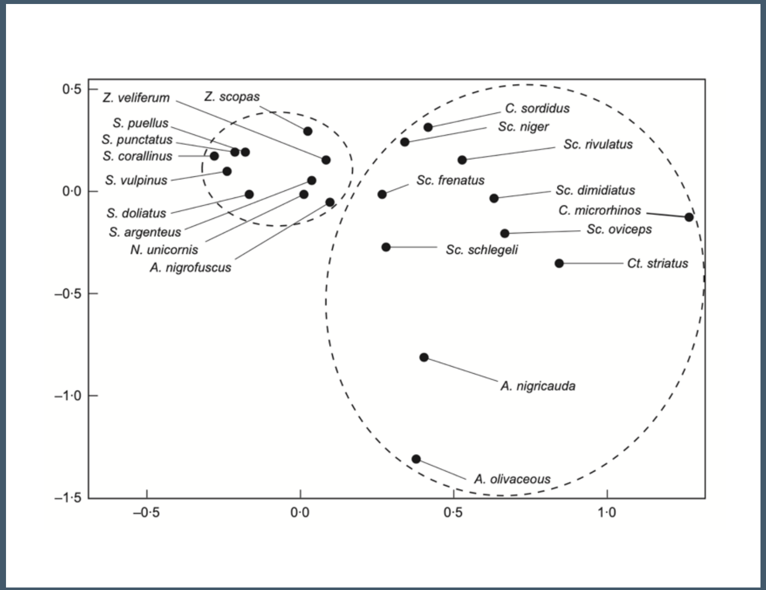
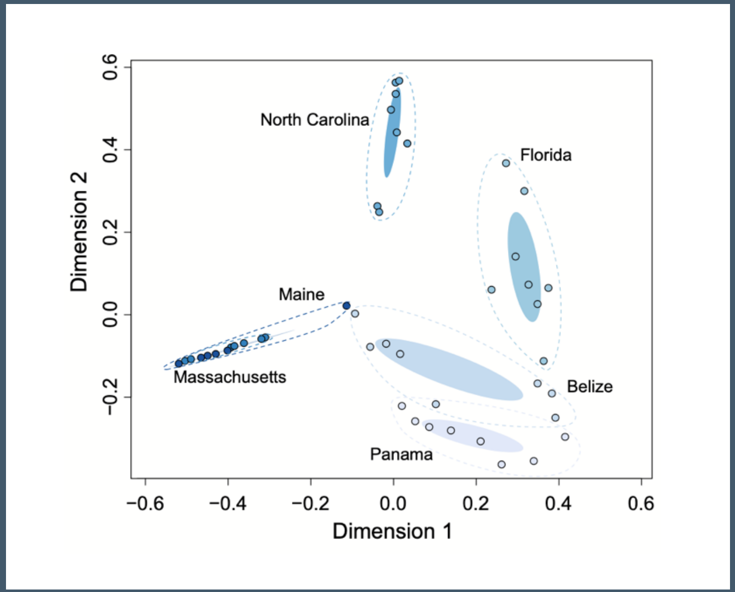
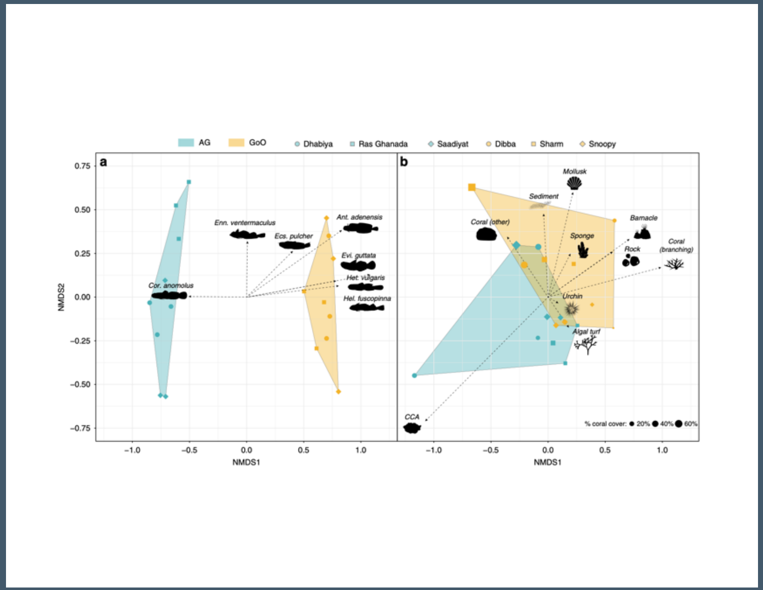
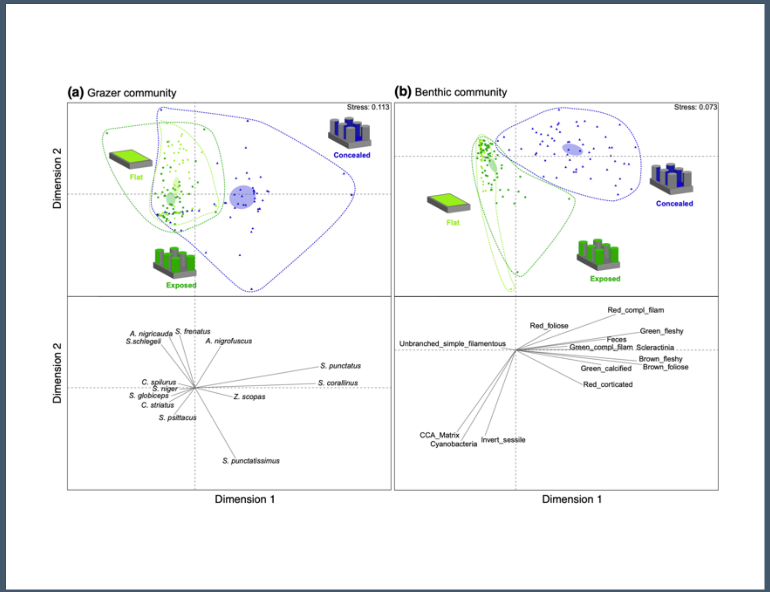

Dimension 2
Dimension 1
Florida Away
Florida Dock
North Carolina Away
North Carolina Dock
Massachusetts Away
Massachusetts Dock
Maine Away
Maine Dock
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6


MDS2
MDS1
Species
A. aspera
A. spinosa
E. altivelis
E. matador
Site
Cormorant
Curlew
Remora
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0


Z. veliferum
Z. scopas
S. puellus
S. punctatus
S. corallinus
S. vulpinus
S. doliatus
S. argenteus
N. unicornis
A. nigrofuscus
C. sordidus
Sc. niger
Sc. rivulatus
Sc. frenatus
Sc. dimidiatus
C. microrhinos
Sc. oviceps
Sc. schlegeli
Ct. striatus
A. nigricauda
A. olivaceous
0·5
0·0
−0·5
−1·0
−1·5
−0·5
0·0
0·5
1·0


Dimension 2
Dimension 1
North Carolina
Florida
Maine
Massachusetts
Belize
Panama
0.6
0.4
0.2
0.0
−0.2
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6


AG
GoO
Dhabiya
Ras Ghanada
Saadiyat
Dibba
Sharm
Snoopy
a
b
NMDS2
NMDS1
NMDS1


(a) Grazer community
(b) Benthic community
Dimension 2
Dimension 2
Dimension 1
Dimension 1


See you Thursday